YEAR 12 MATHEMATICS EXTENSION 1

Year 11 Chapter 14 Combinatorics

Lesson and Homework Pack

Revision	Probability basics	Advanced · Year 11
Lesson 14A	Factorial notation	Past papers: 9 marks · 15 min
Lesson 14B	Counting ordered selections	Past papers: 13 marks · 20 min
Lesson 14C	Ordered selections and grouping	Past papers: 12 marks · 20 min
Lesson 14D	Ordered selections with identical elements	Past papers: 16 marks · 25 min
Lesson 14E	Counting unordered selections	Past papers: 20 marks · 30 min
Lesson 14F	Using counting in probability	Past papers: 15 marks · 25 min
Review	Chapter 14 review exercise	Mixed practice

Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. Answers and full past-paper solutions are in the separate Homework Answers booklet.

CHAPTER REVISION

Probability basics

A short review of Advanced Year 11 probability (Chapter 12): sample spaces, equally likely outcomes, complements, the words “and”, “or” and “not”, and the product rule. Lesson 14F uses all of these, and the counting methods in 14A–14E are what let us find the sizes of the sample space and the event.

SUMMARY · keep this page open while you work

SAMPLE SPACES AND EQUALLY LIKELY OUTCOMES (12A)

- A **sample space** S is the set of possible outcomes of an experiment, where one and only one outcome occurs. It is **uniform** when all outcomes are equally likely (a fair die, a fair coin, a card chosen at random).
- An **event** E is a subset of S. Its outcomes are called **favourable**, and the set of favourable outcomes is the **event space**.

$$P(E) = \frac{|E|}{|S|} = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}$$

A standard pack has 52 cards in four suits (clubs ♣ and spades ♠ are black; diamonds ♦ and hearts ♥ are red). Each suit has 13 cards: A, 2, 3, ..., 10, J, Q, K. The 12 jacks, queens and kings are the **court cards**.

COMPLEMENTARY EVENTS: THE WORD “NOT” (12A)

The complement $\overline{E}$ is the event “E does not occur”, and $P(\overline{E}) = 1 - P(E)$. Consider the complement whenever a question says **not** or **at least one**: “at least one” is the complement of “none”.

SETS, “AND” AND “OR” (12C, 12D)

“A and B” is the intersection $A \cap B$. “A or B” is the union $A \cup B$, and “or” always means “and/or”. For any two events, the **addition rule** is

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

- It comes from counting sets: $|A \cup B| = |A| + |B| - |A \cap B|$.
- Events are **mutually exclusive** (disjoint) when they cannot both occur. Then $P(A \text{ or } B) = P(A) + P(B)$.

MULTI-STAGE EXPERIMENTS AND THE PRODUCT RULE (12E)

- If A and B are events in successive **independent** stages, then $P(AB) = P(A) \times P(B)$. Independent means that one stage does not affect the probabilities of the other, as with repeated tosses of a coin.
- For a selection **without replacement** the stages are not independent: after each draw the numbers left in the bag change, so use the new totals for the next stage.
- The same idea counts outcomes. If one stage can be done in m ways and the next in n ways, the two stages can be done in $m \times n$ ways. This multiplication principle is the start of Chapter 14.

WARM-UP · answers are in the answers booklet

1 A fair die is thrown once. Let E be the event "the number is even or a multiple of 3".

a Write down the sample space S and the event space E.

b Find $P(E)$.

c Find the probability that the number is not in E.

2 A card is drawn at random from a standard pack of 52 cards. Find the probability that it is

a a heart,

b a court card,

c not a court card,

d a red card or a king.

3 In a class of 30 students, 18 play soccer, 14 play tennis and 6 play both sports. A student is chosen at random. Find the probability that the student plays

a neither sport,

b exactly one of the two sports.

4 A fair coin is tossed four times.

a How many outcomes are in the sample space?

b Find the probability that the first two tosses are heads.

c Find the probability that there is at least one tail.

5 A bag holds 5 red and 3 blue discs. Two discs are drawn in succession without replacement. Find the probability that

a both discs are red,

b the two discs are different colours,

c at least one disc is blue.

LESSON 14A

Factorial notation

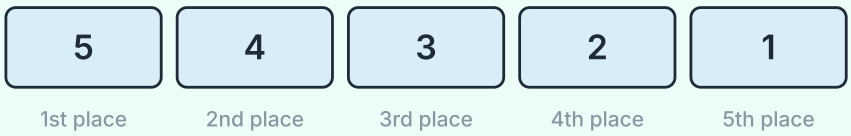
Counting problems keep producing products of descending whole numbers, such as $5 \times 4 \times 3 \times 2 \times 1$. This lesson gives these products a name and a notation, $n!$, defines them informally and recursively, explains why $0! = 1$, and shows how **unrolling** a factorial simplifies quotients, sums and differences of factorials.

CONCEPT 1 · the definition of n factorial

WHERE FACTORIALS COME FROM

How many ways can five people form a queue at a bus stop? Fill the places one at a time.

- Choose the person at the head of the queue in five ways.
- Four people remain, so the person in second place can be chosen in four ways.
- Three people remain, so the person in third place can be chosen in three ways.
- Two people remain, so the person in fourth place can be chosen in two ways.
- And there is now only one way to choose the person in the last place.



So the number of possible queues is $5 \times 4 \times 3 \times 2 \times 1$. The same product appears when $y = x^5$ is differentiated five times: $y^{(5)} = 5 \times 4 \times 3 \times 2 \times 1$. This product is written $5!$ and is called **5 factorial**:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 \qquad (\text{which evaluates to } 120).$$

BOX 1 THE DEFINITION OF n FACTORIAL

- Informally, for each whole number n, the number $n!$ (n *factorial*) is the product of all positive whole numbers from n down to 1:
  $$n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1.$$
- The better definition of the function $n!$ is **recursive**:
  $$\begin{cases} 0! = 1, \\ n! = n \times (n-1)!, \quad \text{for all whole numbers } n \geq 1. \end{cases}$$

The recursive definition avoids the dots in the first formula, it gives $0!$ a meaning, and it gives a better insight into how to manipulate factorial notation.

WHY $0! = 1$, AND THE FIRST FEW FACTORIALS

- An **empty product** is regarded as being 1. If nothing has yet been multiplied, the register remains at 1, where it was originally set in preparation for multiplication.
- In the same way, an **empty sum** is 0: if nothing has yet been added, the register remains at 0.

Using the recursive definition:

$0! = 1$
$1! = 1 \times 0! = 1$
$2! = 2 \times 1! = 2$
$3! = 3 \times 2! = 6$
$4! = 4 \times 3! = 24$
$5! = 5 \times 4! = 120$
$6! = 6 \times 5! = 720$
$7! = 7 \times 6! = 5040$
$8! = 8 \times 7! = 40\,320$
$9! = 9 \times 8! = 362\,880$
$10! = 10 \times 9! = 3\,628\,800$
$11! = 11 \times 10! = 39\,916\,800$

Factorials increase very quickly indeed. Calculators have a factorial button labelled $\boxed{x!}$ or $\boxed{n!}$. It confirms that $0! = 1$, and it gives an error message if n is not a whole number: the domain of the function $n!$ is the whole numbers $0, 1, 2, \ldots$

TUTOR DEMO **Queues, derivatives and the recursive definition**

a In how many ways can six people form a queue? Write the answer in factorial notation, then evaluate it.
b Differentiate $y = x^4$ four times, and show that $y^{(4)} = 4!$.
c Given that $9! = 362\,880$, use the recursive definition to find $10!$ and $8!$.

YOUR TURN **Same skills**

a Seven runners finish a race, with no ties. In how many orders can they finish? Give the answer in factorial notation and as a number.
b Differentiate $y = x^3$ three times, and show that $y''' = 3!$.
c Given that $7! = 5040$, use the recursive definition to find $8!$ and $6!$.

WORKED EXAMPLE **From the text: keep for homework**

a How many ways can five people form a queue at a bus stop?
b What is the fifth derivative of $y = x^5$?

SOLUTION

a The first place can be filled in 5 ways, the second in 4 ways, the third in 3 ways, the fourth in 2 ways and the last in 1 way. So

$$\text{number of queues} = 5 \times 4 \times 3 \times 2 \times 1 = 5! = 120.$$

b Differentiating five times,

$$\begin{aligned} y' &= 5x^4 \\ y'' &= 5 \times 4 \times x^3 \\ y''' &= 5 \times 4 \times 3 \times x^2 \\ y^{(4)} &= 5 \times 4 \times 3 \times 2 \times x \\ y^{(5)} &= 5 \times 4 \times 3 \times 2 \times 1 \\ &= 5! = 120. \end{aligned}$$

CONCEPT 2 · unrolling factorials

UNROLLING

The recursive definition is the key idea in many calculations. Successive applications of it can be thought of as **unrolling** the factorial further and further:

$$\begin{aligned} 8! &= 8 \times 7! && (\text{unrolling once}) \\ &= 8 \times 7 \times 6! && (\text{unrolling twice}) \\ &= 8 \times 7 \times 6 \times 5! && (\text{unrolling three times}) \end{aligned}$$

and so on. This idea is vital when there are fractions involved.

BOX 2 CALCULATING BY UNROLLING FACTORIALS

- A factorial can be unrolled one step at a time:
  $7! = 7 \times 6! = 7 \times 6 \times 5! = 7 \times 6 \times 5 \times 4! = \cdots$
- Cancel fractions with factorials by unrolling until the factorials cancel:
  $$\frac{7!}{4!} = \frac{7 \times 6 \times 5 \times 4!}{4!} = 7 \times 6 \times 5 = 210.$$

Unroll the **larger** factorial down to the smaller one. With pronumerals,
$(n+2)! = (n+2)(n+1)n!$ and $n! = n(n-1)!$.

TUTOR DEMO **Cancelling by unrolling**

Simplify each expression using unrolling techniques.

a $\dfrac{9!}{6!}$

b $\dfrac{(n+3)!}{n!}$

c $\dfrac{n!}{(n-4)!}$

d $\dfrac{8!}{5! \times 3!}$

YOUR TURN **Same method**

Simplify each expression using unrolling techniques.

a $\dfrac{11!}{8!}$

b $\dfrac{(n+4)!}{(n+1)!}$

c $\dfrac{n!}{(n-3)!}$

d $\dfrac{9!}{6! \times 3!}$

WORKED EXAMPLE 1 **Keep for homework**

Simplify each expression using unrolling techniques:

a $\dfrac{10!}{7!}$ b $\dfrac{(n+2)!}{(n-1)!}$ c $\dfrac{n!}{(n-r)!}$

SOLUTION

a

$$\begin{aligned}\frac{10!}{7!} &= \frac{10\times 9\times 8\times 7!}{7!}\\ &= 10\times 9\times 8\\ &= 720\end{aligned}$$

b

$$\begin{aligned}\frac{(n+2)!}{(n-1)!} &= \frac{(n+2)(n+1)n(n-1)!}{(n-1)!}\\ &= (n+2)(n+1)n\end{aligned}$$

c

$$\begin{aligned}\frac{n!}{(n-r)!} &= \frac{n(n-1)(n-2)\cdots(n-r+1)(n-r)!}{(n-r)!}\\ &= \underbrace{n(n-1)(n-2)\cdots(n-r+1)}_{r \text{ factors}}\end{aligned}$$

CONCEPT 3 · common denominators and common factors

SUMS AND DIFFERENCES OF FACTORIALS

- To add or subtract fractions with factorials in the denominators, use the **largest factorial** as the common denominator. Unroll it to see what each numerator must be multiplied by:
  $$\frac{1}{8!} = \frac{10\times 9}{10\times 9\times 8!} = \frac{10\times 9}{10!} \quad \text{and} \quad \frac{1}{n!} = \frac{n+1}{(n+1)!}.$$
- Unrolling also lets you **take out a common factor** from a sum or difference of factorials. Unroll each term down to the smallest factorial:
  $$8! - 7! = 8\times 7! - 7! = (8-1)\times 7! = 7\times 7!.$$

TUTOR DEMO **Common denominators, then a common factor**

Simplify each expression using a common denominator.

a $\frac{1}{7!} - \frac{1}{9!}$

b $\frac{1}{(n+1)!} + \frac{1}{(n+2)!}$

Simplify each expression by taking out a common factor.

c $10! - 9!$

d $(n+2)! - (n+1)!$

YOUR TURN **Same methods**

Simplify each expression using a common denominator.

a $\frac{1}{6!} + \frac{1}{8!}$

b $\frac{1}{(n-1)!} - \frac{1}{n!}$

Simplify each expression by taking out a common factor.

c $7! + 5!$

d $n! + (n-1)!$

WORKED EXAMPLE 2 **Keep for homework**

Simplify each expression using a common denominator.

a $\frac{1}{8!} - \frac{1}{10!}$

b $\frac{1}{n!} + \frac{1}{(n+1)!}$

SOLUTION

a

$$\frac{1}{8!} - \frac{1}{10!} = \frac{10 \times 9}{10!} - \frac{1}{10!}$$
$$= \frac{89}{10!}$$

b

$$\frac{1}{n!} + \frac{1}{(n+1)!} = \frac{n+1}{(n+1)!} + \frac{1}{(n+1)!}$$
$$= \frac{n+2}{(n+1)!}$$

TEXTBOOK HOMEWORK · Exercise 14A

FOUNDATION

1 Use arguments similar to the start of this section to express the following in factorial notation.

a The number of ways to form a queue with 8 people.

b The number of ways to arrange 4 distinct books on a bookshelf.

c The number of ways 3 particular people can be placed first, second and third in a competition, assuming that each is placed in one of these three positions.

d Sam drops his portable keyboard and all 101 keys fall off. How many ways can he put the keys back on the keyboard, assuming that they are all interchangeable?

e When the teacher calls on students in alphabetical order, Andrzej Zywiec is upset that he is always called last. In a class of 20, how many ways could the roll be called, if the alphabetical ordering restriction were dropped?

2 Use the definition of $n!$ and methods of unrolling factorials to evaluate each expression. Do not use a calculator.

a $3!$

b $5!$

c $1!$

d $\dfrac{15!}{14!}$

e $\dfrac{10!}{8! \times 2!}$

f $\dfrac{15!}{13! \times 3!}$

g $\dfrac{12!}{3! \times 9!}$

h $\dfrac{8!}{4! \times 4!}$

3 Use the factorial button $\boxed{n!}$ on your calculator to evaluate each expression.

a $7!$

b $10!$

c $0!$

d $\dfrac{9!}{4!}$

e $\dfrac{8!}{3!}$

f $\dfrac{10!}{5! \times 3! \times 2!}$

g $\dfrac{15!}{3! \times 5! \times 9!}$

h $\dfrac{12!}{2! \times 3! \times 4! \times 5!}$

4 If $f(x) = x^6$, find expressions for:

a $f'(x)$

b $f''(x)$

c $f'''(x)$

d $f''''(x)$

e $f^{(5)}(x)$

f $f^{(6)}(x)$

g $f^{(7)}(x)$

5 Simplify by unrolling factorials appropriately:

a $\dfrac{n!}{(n-1)!}$

b $n \times (n-1)!$

c $\dfrac{n(n-1)!}{n!}$

d $\dfrac{(n+1)!}{(n-1)!}$

e $\dfrac{(n+2)!}{n!}$

f $\dfrac{(n-2)!}{n!}$

g $\dfrac{(n-2)!\,(n-1)!}{n!\,(n-3)!}$

h $\dfrac{n!\,(n-1)!}{(n+1)!}$

6 Simplify by taking out a common factor:

a $8! - 7!$

b $(n+1)! - n!$

c $8! + 6!$

d $(n+1)! + (n-1)!$

e $9! + 8! + 7!$

f $(n+1)! + n! + (n-1)!$

DEVELOPMENT

7 Write each expression as a single fraction.

a $\dfrac{1}{n!} + \dfrac{1}{(n-1)!}$

b $\dfrac{1}{n!} - \dfrac{1}{(n+1)!}$

c $\dfrac{1}{(n+1)!} - \dfrac{1}{(n-1)!}$

8 a If $f(x) = x^n$, find:

i $f'(x)$

ii $f''(x)$

iii $f^{(n)}(x)$

iv $f^{(k)}(x)$, where $k \leq n$.

b If $f(x) = \dfrac{1}{x}$, find:

i $f'(x)$

ii $f''(x)$

iii $f^{(5)}(x)$

iv $f^{(n)}(x)$.

9 a Show that $k \times k! = (k+1)! - k!$

b Hence by considering each individual term as a difference of two terms, sum the series $1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n!$

10 a Find the largest power of: i 2, ii 10, that is a divisor of $10!$

b Find the largest power of: i 2, ii 5, iii 7, iv 13, that is a divisor of $100!$

11 [A relationship between higher derivatives of polynomials and factorials]

a If $f(x) = 11x^3 + 7x^2 + 5x + 3$, show that:

i $f(0) = 3 \times 0!$ ii $f'(0) = 5 \times 1!$

iii $f''(0) = 7 \times 2!$ iv $f'''(0) = 11 \times 3!$

v $f^{(k)}(0) = 0$, for all $k \geq 4$.

Hence explain why $f(x)$ can be written

$$f(x) = \frac{f(0)}{0!} + \frac{f'(0)\,x}{1!} + \frac{f''(0)\,x^2}{2!} + \frac{f'''(0)\,x^3}{3!}.$$

b Show that if $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ is any polynomial, then

$$f^{(k)}(0) = \begin{cases} a_k\,k!, & \text{for } k = 0, 1, 2, \ldots, n, \\ 0, & \text{for } k > n, \end{cases}$$

and hence explain why $f(x) = \dfrac{f(0)}{0!} + \dfrac{f'(0)\,x}{1!} + \dfrac{f''(0)\,x^2}{2!} + \dfrac{f'''(0)\,x^3}{3!} + \cdots$.

Because $f(x)$ is a polynomial, this *power series* has only finitely many terms. There are ways of generalising this to some other functions that are not polynomials.

12 a Evaluate $\dfrac{k}{(k+1)!}$, for $k = 1,\ 2,\ 3,\ 4$ and 5.

b Evaluate $\dfrac{1}{(1+1)!} + \dfrac{2}{(2+1)!} + \cdots + \dfrac{n}{(n+1)!}$, for $n = 1,\ 2,\ 3,\ 4$ and 5.

c Write the expression in part **b** as S_n, to indicate that it is a sum whose value depends on n. That is,

$$S_n = \frac{1}{(1+1)!} + \frac{2}{(2+1)!} + \cdots + \frac{n}{(n+1)!}.$$

Make a reasonable guess about the value of S_n. Hence find $\lim\limits_{n\to\infty} S_n$.

d Prove that $\dfrac{k}{(k+1)!} = \dfrac{1}{k!} - \dfrac{1}{(k+1)!}$, and hence produce a proof of part **c** using a collapsing sum.

ENRICHMENT

13 Express using factorial notation:

a $30 \times 28 \times 26 \times \cdots \times 2$ b $29 \times 27 \times 25 \times \cdots \times 1$ c $\dfrac{30 \times 28 \times 26 \times \cdots \times 2}{29 \times 27 \times 25 \times \cdots \times 1}$

14 [Stirling's formula] The following formula is too difficult to prove at this stage, but it is most important because it provides a continuous function that approximates $n!$ for integer values of n:

$$n! \doteq \sqrt{2\pi}\, n^{n+\frac{1}{2}}\, e^{-n}, \quad \text{in the sense that the percentage error} \to 0 \text{ as } n \to \infty.$$

Show that the formula has an error of approximately 2.73% for $3!$ and 0.83% for $10!$ Find the percentage error for $60!$

PAST-PAPER HOMEWORK · mini exam

Suggested time: 15 minutes **9 marks** Calculator allowed · show working · hints are below each question if you get stuck

Trial papers rarely test factorial notation on its own, so this mini exam is shorter than usual. Questions 1 and 2 use the "fill the places one at a time" argument from the start of the lesson.

PP 1 **Cherrybrook Technology High School, 2020, Year 12 AP4, Mathematics Extension 1, Q1** 1 mark

A PIN is to be chosen using four of the digits 0 through 9. Digits cannot be repeated. How many possible PINs are there?

(A) 256 (B) 210

(C) 10 000 (D) 5040

Hint: does the order of the digits in a PIN matter?

PP 2 **Ascham School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(e)(i)** 1 mark

Find the number of ways in which the letters of the word EPSILON can be arranged in a straight line so that:

i) the arrangement begins with P and ends with S. 1

Hint: which places in the arrangement are already decided?

PP 3 **Normanhurst Boys High School, 2022, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(a)(i)** 1 mark

Show that

$$\frac{1}{(n+1)!} - \frac{n+1}{(n+2)!} = \frac{1}{(n+2)!}$$

Hint: how are $(n+1)!$ and $(n+2)!$ related?

PP 4 **Normanhurst Boys High School, 2021, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(e)(i)** 1 mark

Show that $\dfrac{1}{(n+2)!} - \dfrac{n+1}{(n+3)!} = \dfrac{2}{(n+3)!}$, where n is an integer.

Hint: the same method as PP 3. Which factorial is the larger one?

PP 5 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q12(c)** 2 marks

Solve $(n+2)! = 72n!$

Hint: unrolling turns this into an ordinary equation in n. Remember what values n is allowed to take.

PP 6 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(d)** 3 marks

Show that $(n+1)\left[n!\,n + (n-1)!\,(2n-1) + (n-2)!\,(n-1)\right] = (n+2)!$

Hint: unroll each factorial down to the smallest one.

LESSON 14B

Counting ordered selections

Counting is the key to many probability problems. Two questions decide the method: **is the selection ordered or unordered?** and, if it is ordered, **is repetition allowed?** This lesson counts ordered selections. It sets out the multiplication principle with box diagrams, counts selections with repetition (n^r) and without repetition (${}^n\mathrm{P}_r$), and shows why the restrictions in a problem must be dealt with first.

CONCEPT 1 · the multiplication principle and box diagrams

ORDERED OR UNORDERED? WITH OR WITHOUT REPETITION?

Probability problems become much easier with a more systematic approach to counting. Two questions dominate:

- Are the selections we are counting **ordered** or **unordered**?
- If they are ordered, is **repetition** allowed or not?

Sections 14B–14D develop the theory of counting ordered selections, with and without repetition. Section 14E deals with unordered selections.

THE MULTIPLICATION PRINCIPLE

This morning, Stefan chose a shirt to wear from the twelve that he owns. Then, without worrying about a match, he chose a pair of trousers from the six pairs he owns. How many possible outfits could he have walked out of the house wearing?

The answer is $12 \times 6 = 72$. Each choice of trousers was part of twelve outfits, depending on the choice of shirt. There were six choices of trousers, so

$$\text{number of outfits} = 12 + 12 + 12 + 12 + 12 + 12 = 12 \times 6.$$

The multiplication principle may be applied to more than two successive choices. Stefan also chose a pair of shoes from his eight pairs, and a tie from his collection of 87 ties. Taking these choices into account,

$$\text{number of outfits} = 12 \times 6 \times 8 \times 87 = 50\,112.$$

BOX 3 THE MULTIPLICATION PRINCIPLE

Suppose that a selection is to be made in r stages. Suppose that the first stage can be chosen in n_1 ways, the second in n_2 ways, ..., the rth in n_r ways. Then

$$\text{number of ways of choosing the complete selection} = n_1 \times n_2 \times \cdots \times n_r.$$

USING BOXES FOR MULTIPLICATION PRINCIPLE QUESTIONS

An ordered selection can usually be regarded as a sequence of choices made one after the other. An efficient setting-out is to use a **box diagram** to keep track of these successive choices. In the situation above with Stefan's outfits:

shirt	trousers	shoes	tie
12	6	8	87

$$\text{Number of outfits} = 12 \times 6 \times 8 \times 87 = 50\,112.$$

Note: in these questions a "word" is any arrangement of letters, whether or not it is in a dictionary. There are 5 vowels (A, E, I, O, U) and 21 consonants. Unless otherwise indicated, always take the letter 'y' as a consonant.

TUTOR DEMO **Successive choices in boxes**

a Ella chooses a shirt from her 10 shirts, a skirt from her 4 skirts, a pair of shoes from her 7 pairs and a hat from her 3 hats. How many outfits are possible?
b How many four-letter words can be formed in which the first and last letters are vowels and the middle two letters are consonants?

YOUR TURN **Same method**

a A café menu has 5 entrées, 8 mains and 4 desserts. How many three-course meals, with one dish from each course, are possible?
b How many three-letter words can be formed in which the first and last letters are consonants and the middle letter is a vowel?

WORKED EXAMPLE 3 **Keep for homework**

How many five-letter words can be formed in which the second and fourth letters are vowels and the other three letters are consonants?

SOLUTION

We can select each letter in order:

1st letter	2nd letter	3rd letter	4th letter	5th letter
21	5	21	5	21

$$\text{Number of words} = 21 \times 5 \times 21 \times 5 \times 21 = 231\,525.$$

CONCEPT 2 · ordered selections with repetition

EACH LETTER CAN BE CHOSEN IN n WAYS

Suppose that r-letter words are to be formed from n distinct letters, where any letter can be used any number of times. Then each successive letter in the word can be chosen in n ways:

1st letter	2nd letter	3rd letter	...	rth letter
n	n	n	...	n

giving n^r distinct words altogether.

BOX 4 ORDERED SELECTIONS WITH REPETITION

The number of r-letter words formed from n distinct letters, with repetition, is n^r.

"AT LEAST ONE": COUNT THE COMPLEMENT

To count the selections containing **at least one** of something, it is usually easier to count those containing **none** of it, and subtract from the total. This is the counting form of the complement rule on the revision page.

TUTOR DEMO **Repetition allowed, then a complement**

a How many five-digit numbers can be formed entirely from the digits 1, 2, 3 and 4?
b How many of these numbers contain at least one 4?

YOUR TURN **Same method**

a How many four-letter words can be formed from the letters A, B, C, D, E and F, if any letter may be used any number of times?
b How many of these words contain at least one A?

WORKED EXAMPLE 4 **Keep for homework**

a How many six-digit numbers can be formed entirely from odd digits?
b How many of these numbers contain at least one seven?

SOLUTION

a There are five odd digits, so the number of such numbers is 5^6.

b We first count the number of these six-digit numbers not containing 7. Such numbers are formed from the digits 1, 3, 5 and 9, so there are 4^6 of them. Subtracting this from the answer to part **a**,

$$\text{number of numbers} = 5^6 - 4^6 = 11\,529.$$

CONCEPT 3 · permutations without repetition

ORDERED SELECTIONS WITHOUT REPETITION

Counting ordered selections without repetition typically involves factorials, because as each stage is completed, the number of objects to choose from diminishes by 1.

Note: the phrases ‘with replacement’ and ‘with repetition’ are used almost interchangeably. Similarly, the phrases ‘without replacement’ and ‘without repetition’ are also used almost interchangeably.

THE GENERAL CASE: PERMUTATIONS

A *permutation* or *ordered set* is an arrangement of objects chosen from a certain set. For example, the words ABC, CED, EAB and DBC are some of the permutations of three letters taken from the 5-member set $\{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}\}$.

The symbol ${}^{n}\mathrm{P}_{r}$ is used to denote the number of permutations of r letters chosen without repetition from a set of n distinct letters. Fill the r places one at a time:

1st letter	2nd letter	3rd letter	4th letter	...	rth letter
n	$n-1$	$n-2$	$n-3$	...	$n-r+1$

$$\begin{aligned}\text{Hence } {}^{n}\mathrm{P}_{r} &= n(n-1)(n-2)(n-3)\cdots(n-r+1) \\ &= \frac{n(n-1)(n-2)(n-3)\times\cdots\times 2\times 1}{(n-r)(n-r-1)\times\cdots\times 2\times 1} \\ &= \frac{n!}{(n-r)!}.\end{aligned}$$

BOX 5 PERMUTATIONS: ORDERED SELECTIONS WITHOUT REPETITION

- A *permutation* is an arrangement of objects chosen from a certain set without repetition, that is, without replacement.
- We often imagine permutations as words formed from distinct letters chosen from a larger word consisting only of distinct letters.
- The number of permutations with r letters that can be formed without repetition from a set of n distinct letters is

$$ {}^{n}\mathrm{P}_{r} = \frac{n!}{(n-r)!}.$$

The phrases ‘without repetition’ and ‘without replacement’ are used almost interchangeably. The term ‘distinct objects’ is also used to describe this situation.

EVALUATING ${}^{n}\mathrm{P}_{r}$, AND THE PERMUTATIONS OF A SET

For example, the number of permutations of three distinct letters taken from the 5-member set $\{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}\}$ is ${}^{5}\mathrm{P}_{3} = \dfrac{5!}{2!} = 60$.

Scientific calculators have a button labelled $\boxed{{}^{n}\mathrm{P}_{r}}$ which will find values of ${}^{n}\mathrm{P}_{r}$. For low values of n and r the answers are exact, but for high values they are only approximations. Make a practice of evaluating these numbers by hand when it is reasonable to do so, because such calculations greatly help the intuition.

A *permutation of a set* is an arrangement of all the members of the set in a particular order.

If the set has n members, then the number of permutations is ${}^{n}\mathrm{P}_{n} = \dfrac{n!}{0!} = n!$

BOX 6 PERMUTATIONS OF A SET

The number of permutations of an n-member set, that is, the number of distinct orderings of the set, is ${}^{n}\mathrm{P}_{n} = n!$

Box 6 summarises the arguments used at the start of Lesson 14A and in Question 1 of Exercise 14A.

TUTOR DEMO **Permutations by boxes and by formula**

a In how many ways can a club of 14 members choose a president, a vice-president and a secretary?
b How many six-digit numbers, and how many seven-digit numbers, can be formed from the digits 1, 2, 3, 4, 5, 6 and 7 if no repetition is allowed?
c In how many ways can 10 people form a queue? Will the number of ways double with 20 people?
d Write ${}^{6}\mathrm{P}_{4}$ as a product, and as a quotient of factorials, then evaluate it by hand.

YOUR TURN **Same skills**

a In how many ways can a class of 25 choose a captain and a vice-captain?
b How many four-digit numbers, and how many five-digit numbers, can be formed from the digits 1, 2, 3, 4 and 5 if no repetition is allowed?
c In how many ways can 6 different books be arranged in a row on a shelf?
d Write ${}^{8}\mathrm{P}_{3}$ as a product, and as a quotient of factorials, then evaluate it by hand.

WORKED EXAMPLE 5 **Keep for homework**

In how many ways can a class of 16 select a committee consisting of a president, a vice-president, a treasurer and a secretary?

SOLUTION

Select in order the president, the vice-president, the treasurer and the secretary (we assume that the same person cannot fill two roles).

president	vice-president	treasurer	secretary
16	15	14	13

Hence there are $16 \times 15 \times 14 \times 13 = \dfrac{16!}{12!}$ possible committees (which evaluates to $43\,680$).

WORKED EXAMPLE 6 **Keep for homework**

How many eight-digit numbers, and how many nine-digit numbers, can be formed from the nine non-zero digits if no repetition is allowed?

SOLUTION

$$\begin{aligned}\text{Number of 8-digit numbers} &= {}^9\mathrm{P}_8 \\ &= \frac{9!}{1!} \\ &= 9!\end{aligned}$$

$$\begin{aligned}\text{Number of 9-digit numbers} &= {}^9\mathrm{P}_9 \\ &= \frac{9!}{0!} \\ &= 9!\end{aligned}$$

WORKED EXAMPLE 7 **Keep for homework**

In how many ways can 20 people form a queue? Will the number of ways double with 40 people?

SOLUTION

With 20 people, number of ways $= 20!$ $(\doteq 2.4 \times 10^{18})$.
With 40 people, number of ways $= 40!$ $(\doteq 8.2 \times 10^{47})$.
Hence 40 people can form a queue in about 3×10^{29} times more ways than 20 people.

CONCEPT 4 · deal with the restrictions first

A GENERAL COUNTING PRINCIPLE

Many problems have some restrictions in the way things can be arranged. These restrictions should be dealt with first. It is also important to keep in mind that the order of the boxes represents the order in which the choices are made, not the final ordering of the objects, and that they can be used in surprisingly flexible ways.

BOX 7 DEAL WITH THE RESTRICTIONS FIRST

- When counting ordered selections, deal with any restrictions first.
- Place the boxes in the order in which the selections are made.

RESTRICTIONS ON DIGITS

When numbers are formed from digits, the usual restrictions are on the first digit (a number cannot begin with 0) and on the last digit (an even number ends in an even digit, and a multiple of 5 ends in 0 or 5). Fill those boxes first, then the unrestricted places.

TUTOR DEMO **Restricted places first**

a Ten people form two queues, each with five people. Alice will only stand in the left-hand queue, Bob will only stand in the right-hand queue, and Carl and Dina insist on standing in the same queue. In how many ways can the two queues be formed?
b How many odd four-digit numbers can be formed from the digits 1, 2, 3, 4, 5, 6 and 7 if no repetition is allowed?
c How many four-digit numbers can be formed from the digits 0, 1, 2, 3, 4 and 5 if no repetition is allowed?

YOUR TURN **Same method**

a Six people form two queues, each with three people. Ann will only stand in the left-hand queue and Ben will only stand in the right-hand queue. In how many ways can the two queues be formed?
b How many even three-digit numbers can be formed from the digits 1, 2, 3, 4, 5 and 6 if no repetition is allowed?
c How many three-digit numbers can be formed from the digits 0, 1, 2, 3 and 4 if no repetition is allowed?

WORKED EXAMPLE 8 **Keep for homework**

Eight people form two queues, each with four people. Albert will only stand in the left-hand queue, Beth will only stand in the right-hand queue, and Charles and Diana insist on standing in the same queue. In how many ways can the two queues be formed?

SOLUTION

Place Albert in any of the 4 possible positions in the left-hand queue.
Then place Beth in any of the 4 positions in the right-hand queue.
Place Charles in any of the remaining 6 positions.
Place Diana in one of the 2 remaining positions in the same queue.
There remain 4 unfilled positions, which can be filled in $4!$ ways:

Albert	Beth	Charles	Diana	Last four positions
4	4	6	2	$4!$

$$\text{Hence number of ways} = 4 \times 4 \times 6 \times 2 \times 4!$$
$$= 4608.$$

TEXTBOOK HOMEWORK · Exercise 14B

FOUNDATION

1 List all the permutations of the letters of the word DOG. How many are there?

2 List all the permutations of the letters EFGHI, beginning with F, taken three at a time.

3 Find how many arrangements of the letters of the word FRIEND are possible if the letters are taken:

a four at a time, **b** six at a time.

4 Find how many four-digit numbers can be formed using the digits 5, 6, 7, 8 and 9 if:

a no digit is to be repeated, **b** any of the digits can occur more than once.

5 How many three-digit numbers can be formed using the digits 2, 3, 4, 5 and 6 if no digit can be repeated? How many of these are greater than 400?

6 In how many ways can seven people be seated in a row of seven different chairs?

7 The symbol ${}^{n}\mathrm{P}_{r}$ is the number of ways of arranging r objects selected without replacement from n objects. Use this expression, and the $\boxed{{}^{n}\mathrm{P}_{r}}$ button on your calculator, to answer the following questions.

a From a group of 10, three people line up to buy tickets. How many ways can this happen?

b Five cards are each labelled uniquely with one of the digits 1, 2, 3, 4, 5. Three of the five cards are placed down in a row. How many ways can the cards be arranged?

c One hundred people each buy one ticket in a lottery. How many ways can the first three places be awarded?

8 The expression n^r is the number of ways of arranging r objects selected with replacement from n objects. Use this expression to answer the following questions.

a How many three-digit numbers can be written down from the digits 1 to 9? The digits need not be distinct.

b One hundred people each buy one ticket in a lottery. Tickets are selected one by one at random from a barrel and then replaced between selections. How many ways can the first three places be awarded?

c A computer sends a string of ten binary digits, that is, each symbol can only be 0 or 1. How many such ten-digit strings are possible?

9 Eight runners are participating in a 400-metre race.

a In how many ways can they finish?

b In how many ways can the gold, silver and bronze medals be awarded?

10 **a** If you toss a coin and roll a die, how many outcomes are possible?

b If you toss two coins and roll three dice, how many outcomes are possible?

11 A woman has four hats, three blouses, five skirts, two handbags and six pairs of shoes. In how many ways can she be attired, assuming that she wears one of each item?

12 Jack has six different football cards and Meg has another eight different football cards. In how many ways can one of Jack's cards be exchanged for one of Meg's cards?

DEVELOPMENT

13 In Sydney, landline phone numbers, ignoring the area code, consist of eight digits. Let us consider those starting with the digit 9.

a How many such phone numbers are possible?

b How many of these end in an odd number?

c How many consist of odd digits only?

d How many are there that do not contain a zero, and in which the consecutive digits alternate between odd and even?

14 Users of automatic teller machines are required to enter a four-digit PIN (personal identification number). Find how many PINs:

a are possible,

b consist of four distinct digits,

c consist of odd digits only,

d start and end with the same digit.

15 **a** If repetitions are not allowed, how many four-digit numbers can be formed from the digits $1, 2, \ldots 8, 9$?

b How many of these end in 1?

c How many of these are even?

d How many are divisible by 5?

e How many are greater than 7000?

16 Repeat the previous question if repetitions are allowed.

17 In Tasmania, a car licence plate consists of two letters followed by four digits. Find how many of these are possible:

a if there are no restrictions,

b if there is no repetition of letters or digits,

c if the second letter is X and the third digit is 3,

d if the letters are D and Q and the digits are 3, 6, 7 and 9.

18 **a** In how many ways can the letters of the word NUMBER be arranged?

b How many begin with N?

c How many begin with N and end with U?

d In how many is the N somewhere to the left of the U?

19 In how many ways can a boat crew of eight women be arranged if three of the women can only row on the bow side and two others can only row on the stroke side?

20 A motor bike can carry three people: the driver, one passenger behind the driver and one in the sidecar. If among five people, only two can drive, in how many ways can the bike be filled?

21 a How many five-digit numbers can be formed from the digits 2, 3, 4, 5 and 6?

b How many of these numbers are greater than $56\,432$?

c How many of these numbers are less than $56\,432$?

22 a Integers are formed from the digits 2, 3, 4 and 5, with repetitions not allowed.

i How many such numbers are there?

ii How many of them are even?

23 a How many five-digit numbers can be formed from the digits 0, 1, 2, 3 and 4 if repetitions are not allowed?

b How many of these are odd?

c How many are divisible by 5?

24 a If ${}^{8}\mathrm{P}_{r} = 336$, find the value of r.

b If $7 \times {}^{2n}\mathrm{P}_{n} = 4 \times {}^{2n+1}\mathrm{P}_{n}$, find n.

c Using the result ${}^{n}\mathrm{P}_{r} = \dfrac{n!}{(n-r)!}$, prove that:

i ${}^{n+1}\mathrm{P}_{r} = {}^{n}\mathrm{P}_{r} + r \times {}^{n}\mathrm{P}_{r-1}$

ii ${}^{n}\mathrm{P}_{r} = {}^{n-2}\mathrm{P}_{r} + 2r \times {}^{n-2}\mathrm{P}_{r-1} + r(r-1) \times {}^{n-2}\mathrm{P}_{r-2}$

ENRICHMENT

25 Recall that a whole number is divisible by 3 if and only if the sum of the digits is divisible by 3.

a How many 5-digit whole numbers are divisible by 3?

b How many 5-digit whole numbers with no 0s are divisible by 3?

c How many 5-digit whole numbers with no 0s are divisible by 2?

d How many 5-digit whole numbers with no 0s are divisible by 6?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 20 minutes **13 marks** Calculator allowed · show working · hints are below each question if you get stuck

Trial papers usually test ordered selections inside longer questions on grouping or probability, so this mini exam is a little shorter than usual. Every question uses the methods of this lesson; PP 4 also uses the probability formula from the revision page.

PP 1 **Normanhurst Boys High School, 2023, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q3** 1 mark

How many numbers greater than 2100 can be formed with the digits 1, 2, 3, 4 if no digit is to be used more than once?

(A) 16 (B) 18

(C) 24 (D) 48

Hint: the restriction is on the leading digits, so deal with them first (Box 7).

PP 2 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q10(b)(ii)** 1 mark

Lily and Ben are two of eight people.

(ii) Find the number of ways that the 8 people can arrange themselves in a straight line so that Lily is somewhere to the left of Ben. 1

Hint: what happens to an arrangement if Lily and Ben swap places?

PP 3 **Gosford High School, 2023, Trial HSC examination, Mathematics Extension 1, Q5** 1 mark

In how many ways can 5 people be selected from a group of 6 and then arranged in a line so that the two oldest people in the selected group are at either end of the line? (Note, no two people are the same age)

A. 720

B. 144

C. 72

D. 36

Hint: treat the selection and the arrangement as two stages of the multiplication principle.

PP 4 **Hurlstone Agricultural High School, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q11(b)** 1 mark

The letters of the word CONSIDER are rearranged at random. What is the probability that an arrangement begins with 'SI'?

Hint: use $P(E) = \dfrac{|E|}{|S|}$ from the revision page, and count both sets with boxes.

PP 5 **Girraween High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q14(c)** 2 marks

How many numbers greater than 6000 can be formed with the digits 1, 4, 5, 7, 8 if no digit is repeated?

Hint: think about how many digits such a number could have.

PP 6 **Ascham School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(e)(iii)** 1 mark

Find the number of ways in which the letters of the word EPSILON can be arranged in a straight line so that:

iii) N is somewhere to the left of L. 1

Hint: the same idea as PP 2.

PP 7 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q12(d)** 3 marks

It is given that

$$\frac{{}^{n}\mathrm{P}_{r-1}}{a} = \frac{{}^{n}\mathrm{P}_{r}}{b} = \frac{{}^{n}\mathrm{P}_{r+1}}{c}$$

Prove that $b^2 = a(b+c)$.

Hint: write each ${}^{n}\mathrm{P}_{r}$ in factorial form (Box 5), and remember how unrolling simplifies a quotient of factorials.

PP 8 **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(c)(i)(ii)** 3 marks

A 4-digit number begins with 1 and has exactly one pair of identical digits (not necessarily adjacent).

(i) How many such numbers can be formed if the repeated digit is 1? 1

(ii) How many such numbers can be formed in total? 2

Hint: deal with the restrictions first. Which digits are allowed in the last three places?

LESSON 14C

Ordered selections and grouping

Lesson 14B gave one important counting principle: **deal with the difficulties first**. This lesson adds three more principles that are often useful in counting: **grouping** (order the groups, then order the individuals within each group), **the complementary situation** (count what is not wanted and subtract), and **using cases** (split the count into cases that do not overlap, or subtract the overlap).

CONCEPT 1 · grouping

A GENERAL COUNTING PRINCIPLE: GROUPING

In some ordering problems, particular members must be grouped together. This produces a *compound ordering*, in which the various groups must first be ordered, and then the individuals ordered within each group.

For example, in Worked example 9 below, four boys and four girls form a queue. A couple (one girl and one boy) stand together, and the other three girls stand together. So the queue is made of five groups:

G B	G G G	B	B	B
couple: 2! orders	3 girls: 3! orders	boy	boy	boy

The five groups can be ordered in $5!$ ways. Then the couple can be ordered within their group in $2!$ ways, and the three girls within their group in $3!$ ways. A boy standing on his own is a group with only one member.

BOX 8 GROUPING

- First order the groups, then order the individuals within each group.
- A group may sometimes consist of a single individual.

TUTOR DEMO **Order the groups, then within the groups**

a Five boys and three girls form a queue at the canteen. There is one couple who want to stand together, the other two girls want to stand together, but the other four boys don't care where they stand. How many acceptable ways are there of forming the queue?
b How many arrangements of the letters of the word CAMPING are possible if the vowels must be together?

YOUR TURN **Same method**

a Three boys and three girls form a queue. There is one couple who want to stand together, the other two girls want to stand together, but the other two boys don't care where they stand. How many acceptable ways are there of forming the queue?
b How many arrangements of the letters of the word TRAIN are possible if the vowels must be together?

WORKED EXAMPLE 9 **Keep for homework**

Four boys and four girls form a queue at the bus stop. There is one couple who want to stand together, the other three girls want to stand together, but the other three boys don't care where they stand. How many acceptable ways are there of forming the queue?

SOLUTION

There are five groups: the couple, the group of three girls, and the three groups each consisting of one individual boy. First order the five groups, then order the couple, and the three girls.

order the 5 groups	order the couple	order the 3 girls
$5!$	$2!$	$3!$

$$\text{Hence number of ways} = 5! \times 2! \times 3! = 1440.$$

CONCEPT 2 · deal with the complementary situation

A GENERAL COUNTING PRINCIPLE: THE COMPLEMENTARY SITUATION

Many probability questions have already been solved using complementary events (see the revision page). The same principle applies to counting: **count the unacceptable orderings, then subtract them from the total number of orderings**. The word 'not' may or may not be there to prompt you.

A common use is with members that must be **apart**. Counting the orderings in which they are apart is hard, but counting the orderings in which they are together is easy, by grouping (Box 8). So subtract the 'together' orderings from the total.

TUTOR DEMO **Subtract the unacceptable orderings**

a How many six-letter words can be formed from the letters A, B, C, D, E, F if the vowels must be separated by at least one consonant?
b In how many ways can nine people stand in a line if two particular people refuse to stand next to each other?

YOUR TURN **Same method**

a How many five-letter words can be formed from the letters A, B, C, D, E if the vowels must be separated by at least one consonant?
b In how many ways can six people sit in a row if two particular people refuse to sit next to each other?

WORKED EXAMPLE 10 **Keep for homework**

How many seven-letter words can be formed from the letters A, B, C, D, E, F, G if the vowels must be separated by at least one consonant?

SOLUTION

$$\text{Number of orderings without restriction} = 7!$$

$$\text{Number of orderings with A and E together} = 2! \times 6!$$

(Order the A and E and glue them together, then order the 6 groups.)

$$\begin{aligned}\text{Hence number of orderings with A and E apart} &= 7! - 2 \times 6! \\ &= 3600.\end{aligned}$$

CONCEPT 3 · using cases

A GENERAL COUNTING PRINCIPLE: USING CASES

Sometimes the fiddly conditions of the problem mean that one set of boxes is not enough to solve it, and separate cases need to be considered. In this situation, the cases should not overlap, or if they do, the number in the overlap needs to be subtracted.

BOX 9 USING CASES IN COUNTING

- The cases should not overlap.
- If the cases do overlap, the overlap needs to be subtracted.

THE SAME PRINCIPLE WITH SETS

This is the same principle as the counting form of the addition rule on the revision page:

$|A \cup B| = |A| + |B| - |A \cap B|$.

When the two cases A and B do not overlap, $|A \cap B| = 0$ and the two counts are simply added.

TUTOR DEMO **Cases, with and without an overlap**

a How many whole numbers less than 1000 are odd and greater than 600, or multiples of 5 and less than 300?
b How many whole numbers less than 1000 are odd and greater than 300, or multiples of 5 and less than 600?
c How many numbers can be written down using each of the digits 2, 3, 5 and 7 at most once, if the result must be at least 500?

YOUR TURN **Same method**

a How many whole numbers less than 1000 are odd and greater than 700, or multiples of 5 and less than 100?
b How many whole numbers less than 1000 are odd and greater than 400, or multiples of 5 and less than 700?
c How many numbers can be written down using each of the digits 3, 5 and 7 at most once, if the result must be at least 60?

WORKED EXAMPLE 11 **Keep for homework**

a How many whole numbers less than 1000 are odd and greater than 500, or multiples of 5 and less than 200?
b How many whole numbers less than 1000 are odd and greater than 200, or multiples of 5 and less than 500?

SOLUTION

a There are 250 odd numbers between 500 and 1000, and there are 40 multiples of 5 less than 200 (including zero). The two cases do not overlap, so the total number is 290.

b There are 400 odd numbers between 200 and 1000, and there are 100 multiples of 5 less than 500 (including zero). But the two cases overlap: from 200 to 500 there are 30 odd numbers that are multiples of 5. Hence

$$\text{Total} = 400 + 100 - 30 = 470.$$

TEXTBOOK HOMEWORK · Exercise 14C

FOUNDATION

1 How many rearrangements are there of the letters in each word, if the vowels must be together?

a BOARDS

b RIO

c QUIT

d TROUNCE

2 How many arrangements of the word MATHS are possible, if:

a the T and H must remain together?

b the TH must remain together and in this order?

3 How many ways can Andrew, Becky, Courtney, Dion and Ellie sit in a row, if Andrew and Becky sit together and Dion and Ellie sit together?

4 In how many ways can three different Maths books, six different Science books and four different English books be placed on a shelf, if the books relating to each subject are to be kept together?

5 A class is asked to determine how many ways the letters of the word SOLAR can be arranged, if the first two positions cannot both be vowels.

a Jack decides to use cases (the word either starts with a vowel or it does not). Use Jack's method to answer the question.

b Jill decides to consider the complementary situation (both first positions are vowels). Use Jill's method to answer the question.

6 A family of two parents and two children are going on a car trip. Only the parents can drive. If the father drives, then the mother will sit in the back seat where she feels safer. The father always sits in a front seat of the car. How many arrangements are possible, if two sit in the front and two in the back?

7 How many numbers can be written down using each digit of 789 at most once, if the result must be at least 80?

DEVELOPMENT

8 Find how many arrangements of the letters of the word UNIFORM are possible:

a if the vowels must occupy the first, middle and last positions,

b if the word must start with U and end with M,

c if all the consonants must be in a group at the end of the word,

d if the M is somewhere to the right of the U.

9 Find how many arrangements of the letters of the word BEHAVING:

a end in NG,

b begin with three vowels,

c have three vowels occurring together.

10 A Maths test is to consist of six questions. In how many ways can it be arranged so that:

a the shortest question is first and the longest question is last,

b the shortest and longest questions are next to one another?

11 In Morse code, letters are formed by a sequence of dashes and dots. How many different letters is it possible to represent if a maximum of ten symbols are used?

12 Four boys and four girls are to sit in a row. Find how many ways this can be done if:

a the boys and girls alternate,

b the boys and girls sit in distinct groups.

13 Five-letter words are formed without repetition from the letters of PHYSICAL.

a How many consist only of consonants?

b How many begin with P and end with S?

c How many begin with a vowel?

d How many contain the letter Y?

e How many have the two vowels occurring next to one another?

f How many have the letter A immediately following the letter L?

14 **a** How many seven-letter words can be formed without repetition from the letters of the word INCLUDE?

b How many of these do not begin with I?

c How many end in L?

d How many have the vowels and consonants alternating?

e How many have the C immediately following the D?

f How many have the letters N and D separated by exactly two letters?

g How many have the letters N and D separated by more than two letters?

15 Repeat parts **a**–**d** of the previous question if repetition is allowed.

16 **a** In how many ways can ten people be arranged in a line:

i without restriction,

ii if one particular person must sit at either end,

iii if two particular people must sit next to one another,

iv if neither of two particular people can sit on either end of the row?

b In how many ways can n people be placed in a row of n chairs:

i if one particular person must be on either end of the row,

ii if two particular people must sit next to one another,

iii if two of them are not permitted to sit at either end?

17 Five boys and four girls form a queue at the cinema. There are two brothers who want to stand together, the remaining three boys wish to stand together, and the four girls don't mind where they stand. In how many ways can the queue be formed?

18 Eight people are to form two queues of four. In how many ways can this be done if:

a there are no restrictions,

b Jim will only stand in the left-hand queue,

c Sean and Liam must stand in the same queue?

19 There are eight swimmers in a race. In how many ways can they finish if there are no dead heats and the swimmer in Lane 2 finishes:

a immediately after the swimmer in Lane 5,

b any time after the swimmer in Lane 5?

20 Five backpackers arrive in a city where there are five youth hostels.

a How many different accommodation arrangements are there if there are no restrictions on where the backpackers stay?

b How many different accommodation arrangements are there if each backpacker stays at a different youth hostel?

c Suppose that two of the backpackers are brother and sister and wish to stay in the same youth hostel. How many different accommodation arrangements are there if the other three can go to any of the other youth hostels?

21 Numbers less than 4000 are formed from the digits 1, 3, 5, 8 and 9, without repetition.

a How many such numbers are there?

b How many of them are odd?

c How many of them are divisible by 5?

d How many of them are divisible by 3?

ENRICHMENT

22 [Derangements]

A *derangement* of n distinct letters is a permutation of them so that no letter appears in its original position. For example, DABC is a derangement of ABCD, but DACB is not. Denote the number of derangements of n letters by $D(n)$.

a By listing all the derangements of A, AB, ABC and ABCD, find the values of $D(1)$, $D(2)$, $D(3)$ and $D(4)$.

b Suppose that we have formed a derangement of the five letters ABCDE. Let the last letter in the derangement be X, and exchange X with E — this puts E back to its original position. Either X is now also in its original position so that three letters are away from their original positions, or X is not in its original position so that four letters are away from their original positions. Hence explain why

$$D(5) = 4 \times D(4) + 4 \times D(3).$$

c Use this formula to evaluate $D(5)$. Then apply the corresponding arguments and formulae to evaluate $D(6)$, $D(7)$ and $D(8)$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 20 minutes **12 marks** Calculator allowed · show working · hints are below each question if you get stuck

Trial papers rarely test grouping on its own, so this mini exam is a little shorter than usual. PP 5 and PP 8 are probability questions: they combine the counting of Lessons 14B and 14C with $P(E) = \frac{|E|}{|S|}$ from the revision page.

PP 1 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q1** 1 mark

In how many different ways can 5 boys and 3 girls stand in a queue if the girls all stand next to each other?

A. 40 320
B. 720
C. 4 320
D. 3 136

Hint: Box 8. How many groups are there?

PP 2 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q3** 1 mark

How many arrangements of the letters of the word 'OLYMPIC' are possible if the C and the L are to be together?

(A) 120
(B) 720
(C) 240
(D) 1440

Hint: letters that must be together form one group.

PP 3 **Ascham School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(e)(ii)** 1 mark

Find the number of ways in which the letters of the word EPSILON can be arranged in a straight line so that:

ii) the three vowels are adjacent to each other. 1

Hint: first pick out the vowels of EPSILON.

PP 4 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q9** 1 mark

In how many different ways can three girls and three boys be seated on a bench for a photograph if the girls must sit apart?

(A) 72
(B) 120
(C) 144
(D) 576

Hint: seat the boys first. Where can the girls then go?

PP 5 **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q10** 1 mark

Three fair dice of different colours are rolled together. What is the probability that the product of the three scores is a perfect square?

(A) $\frac{6}{216}$

(B) $\frac{13}{216}$

(C) $\frac{32}{216}$

(D) $\frac{38}{216}$

Hint: which of the scores 1 to 6 are perfect squares themselves?

PP 6 **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q11(d)** 2 marks

Donald, Melania and five other people get on a bus one at a time.
In how many ways can the seven people board the bus if Melania refuses to get on immediately following Donald?

Hint: count the complementary situation.

PP 7 **Pymble Ladies' College, 2024, HSC Trial Examination Term 3, Mathematics Extension 1, Q11(b)** 2 marks

Mrs Munro needs to decide the order in which to schedule 8 exams.
Two of these exams are Mathematics Extension 1 and Mathematics Extension 2.
Find the number of different ways that Mrs Munro can schedule the 8 exams so that Mathematics Extension 1 and Mathematics Extension 2 are NOT consecutive.

Hint: the word NOT is a prompt (Concept 2).

PP 8 **Blacktown Boys' High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(c)(i)(ii)** 3 marks

A drawer contains 11 different batteries of which 7 are good and 4 are defective. A teacher selects two batteries for the remote control of the projector and then selects another two for the remote control of the air conditioner. If both batteries must be good for each remote control to work, find the probability that:

(i) Both remote controls work. 1

(ii) Only the remote control for the projector works. 2

Hint: start with part (i): what must be true of all four batteries that the teacher selects?

LESSON 14D

Ordered selections with identical elements

So far, our ordered sets have not permitted any element to be repeated. This lesson deals with permutations of all the letters of a word containing several copies of the one letter. Swapping identical letters does not change a word, so we must allow for **overcounting**.

CONCEPT 1 · counting with identical elements

COUNTING WITH IDENTICAL ELEMENTS

Finding the number of different words formed using all the letters of the word 'PRESSES' is complicated by the fact that there are three Ss and two Es. If the seven letters were all different, we would conclude that

$$\text{number of ways} = 7!$$

But we have *overcounted by a factor of* $2! = 2$, because the Es can be interchanged without changing the word. We have also *overcounted by a factor of* $3! = 6$, because the three Ss can be permuted amongst themselves in $3!$ ways without changing the word. Taking account of both overcountings,

$$\text{number of ways} = \frac{7!}{2! \times 3!} = 420.$$

This method is easily generalised. In the language of 'words':

BOX 10 COUNTING WITH IDENTICAL ELEMENTS

Suppose that a word of n letters has r_1 alike of one type, r_2 alike of another type, ..., r_k alike of a final type. Then the number of distinct words that can be formed by rearranging the letters is

$$\text{number of words} = \frac{n!}{r_1! \times r_2! \times \cdots \times r_k!}.$$

If $r_i = 1$, that is, there is only one letter of the ith type, then $r_i! = 1$, so it doesn't matter whether we include it in the formula or not.

TUTOR DEMO **Divide by the orderings of the identical items**

Four identical mugs and five identical cups are to be arranged in a row along a shelf.
a In how many ways can this be done (counting only the patterns)?
b How does this change if one of the mugs becomes clearly chipped, and two cups break and are replaced by two identical cups different from the other three?
c How many different words can be formed using all the letters of the word PEPPER?

YOUR TURN **Same method**

Two identical vases and four identical jars are to be arranged in a row along a shelf.
a In how many ways can this be done (counting only the patterns)?
b How does this change if one of the vases becomes clearly chipped, and two jars break and are replaced by two identical jars different from the other two?
c How many different words can be formed using all the letters of the word TOOTH?

WORKED EXAMPLE 12 **Keep for homework**

Three identical wine glasses and five identical tumblers are to be arranged in a row across the front of a cupboard.
a In how many ways can this be done (counting only the patterns)?
b How does this change if one of the wine glasses becomes clearly chipped, and two tumblers break and are replaced by two identical tumblers different from the other three?

SOLUTION

a There are 8 glasses, 3 alike of one type and 5 alike of another, so

$$\text{number of ways} = \frac{8!}{3! \times 5!} = 56.$$

b There are now 2 alike of one type, 1 of another, 3 of another, 2 of another, so

$$\text{number of ways} = \frac{8!}{2! \times 1! \times 3! \times 2!} = 1680 \quad (\text{the } 1! \text{ can be omitted}).$$

CONCEPT 2 · words containing only two different letters

WORDS CONTAINING ONLY TWO DIFFERENT LETTERS

The theory about counting with identical letters assumes particular importance when there are only two types of letters, as in part **a** of Worked example 12. The answer to part **a** was $\dfrac{8!}{3! \times 5!}$. This formula is true in general:

BOX 11 PERMUTATIONS OF WORDS CONTAINING ONLY TWO DIFFERENT LETTERS

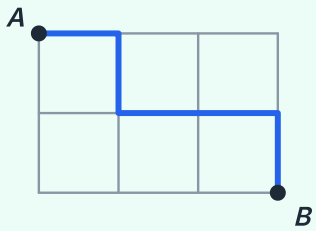
Suppose that an n-letter word consists of r Ys and $n - r$ Ns. Then the number of possible distinct words is

$$\text{number of words} = \frac{n!}{r! \times (n-r)!}.$$

AN APPLICATION: PATHS ON A GRID

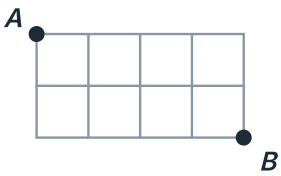
A path from A to B along the grid lines, moving only right or down, is a list of single steps. On the 3 by 2 grid of unit squares shown, every path has 3 steps right (R) and 2 steps down (D), so each path is a five-letter word with three Rs and two Ds. The path drawn is RDRRD. Each such word gives exactly one path, so by Box 11

$$\text{number of paths} = \frac{5!}{3! \times 2!} = 10.$$



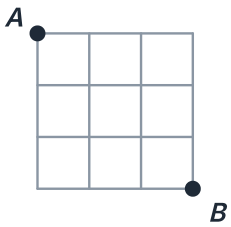
TUTOR DEMO Two types of letters

a Five women and three men form a queue at the bus stop. How many patterns of men and women are there?
b There is space at the bus stop for a queue of eight adults. How many patterns of men and women are possible?
c The diagram shows a grid of unit squares. How many paths are there from A to B moving only right or down along the grid lines?



YOUR TURN **Same method**

a Four women and two men form a queue at the bus stop. How many patterns of men and women are there?
b There is space at the bus stop for a queue of six adults. How many patterns of men and women are possible?
c The diagram shows a grid of unit squares. How many paths are there from A to B moving only right or down along the grid lines?



WORKED EXAMPLE 13 **Keep for homework**

a Six women and four men form a queue at the bus stop. How many patterns of men and women are there?
b There is space at the bus stop for a queue of ten adults. How many patterns of men and women are possible?

SOLUTION

a We can use the formula above, or we can think of arranging ten letters, six alike of one type and four of another.

$$\text{Number of patterns} = \frac{10!}{6! \times 4!} = 210.$$

b Each of the 10 positions can be filled in two possible ways:

1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th
2	2	2	2	2	2	2	2	2	2

$$\text{Hence number of ways} = 2^{10} = 1024.$$

CONCEPT 3 · using cases

USING CASES

As mentioned in Lesson 14C, many counting problems are too complicated to be analysed completely by a single box diagram. In such situations, the use of cases is unavoidable. Attention should be given, however, to minimising the number of different cases that need to be considered.

TUTOR DEMO **One case for each letter left out**

How many six-letter words can be formed by using the letters of the word ‘ADDRESS’?

YOUR TURN **Same method**

How many five-letter words can be formed by using the letters of the word ‘LETTER’?

WORKED EXAMPLE 14 **Keep for homework**

How many six-letter words can be formed by using the letters of the word ‘PRESSES’?

SOLUTION

We omit in turn each of the four letters ‘P’, ‘R’, ‘E’ and ‘S’. This leaves six letters which we must then arrange in order.

1 If an S is omitted, there are then 2 Es and 2 Ss, so

$$\text{number of words} = \frac{6!}{2! \times 2!} = 180.$$

2 If an E is omitted, there are then 3 Ss, so

$$\text{number of words} = \frac{6!}{3!} = 120.$$

3 If P or R is omitted (2 cases), there are then 2 Es and 3 Ss, so

$$\text{number of words} = \frac{6!}{3! \times 2!} \times 2 = 120.$$

(doubling for the two cases)

$$\text{Hence the total number of words is } 180 + 120 + 120 = 420.$$

TEXTBOOK HOMEWORK · Exercise 14D

FOUNDATION

1 Find the number of permutations of the following words if all the letters are used.

a BOB

b ALAN

c SNEEZE

d TASMANIA

e BEGINNER

f FOOTBALLS

g EQUILATERAL

h COMMITTEE

i WOOLLOOMOOLOO

2 The six digits 1, 1, 1, 2, 2, 3 are used to form a six-digit number. How many numbers can be formed?

3 Six coins are lined up on a table. Find how many patterns are possible if there are:

a five tails and one head,

b four heads and two tails,

c three tails and three heads.

4 Eight balls, identical except for colour, are arranged in a line. Find how many different arrangements are possible if:

a all balls are of a different colour,

b there are seven red balls and one white ball,

c there are six red balls, one white ball and one black ball,

d there are three red balls, three white balls and two black balls.

5 Five identical green chairs and three identical red chairs are arranged in a row. Find how many arrangements are possible:

a if there are no restrictions,

b if there must be a green chair on either end.

DEVELOPMENT

6 A motorist travels through eight sets of traffic lights, each of which is red or green. He is forced to stop at three sets of lights.

a In how many ways could this happen?

b What other number of red lights would give an identical answer to part **a**?

7 In how many ways can the letters of the word SOCKS be arranged in a line:

a without restriction,

b so that the two Ss are together,

c so that the two Ss are separated by at least one other letter,

d so that the K is somewhere to the left of the C?

8 **a** Find the number of arrangements of the letters in SLOOPS if:

i there are no restrictions,

ii the two Os are together,

iii the two Os are to be separated,

iv the Os are together and the Ss are together.

b In how many arrangements of the letters in TATTOO are the two Os separated?

9 In how many ways can the letters of the word DECISIONS be arranged:

a without restriction,

b so that the vowels and consonants alternate,

c so that the vowels come first followed by the consonants,

d so that the N is somewhere to the right of the D?

10 In how many ways can the letters of the word PROPORTIONALITY be arranged so that the vowels and consonants still occupy the same places?

11 A form has ten questions in order, each of which requires the answer 'Yes' or 'No'. Find the number of ways the form can be filled in:

a without restriction,

b if the first and last answers are 'Yes',

c if two are 'Yes' and eight are 'No',

d if five are 'Yes' and five are 'No',

e if more than seven answers are 'Yes',

f if an odd number of answers are 'Yes',

g if exactly three answers are 'Yes', and they occur together,

h if the first and last answers are 'Yes' and exactly four more are 'Yes'.

12 Containers are identified by a row of coloured dots on their lids. The colours used are yellow, green and purple. In any arrangement, there are to be no more than three yellow dots, no more than two green dots and no more than one purple dot.

a If six dots are used, what is the number of possible codes?

b What is the number of different codes possible if only five dots are used?

13 **a** How many five-letter words can be formed by using the letters of the word STRESS?

b How many five-letter words can be formed by using the letters of the word BANANA?

14 Find how many arrangements of the letters of the word TRANSITION are possible if:

a there are no restrictions,

b the Is are together,

c the Is are together, and so are the Ns, and so are the Ts,

d the Ns occupy the end positions,

e an N occupies the first but not the last position,

f the letter N is not at either end,

g the vowels are together.

15 Ten coloured marbles are placed in a row.

a If they are all of different colours, how many arrangements are possible?

b What is the minimum number of colours needed to guarantee at least 10 000 different patterns? (Hint: This will need a guess-and-check approach.)

ENRICHMENT

16 Find how many arrangements there are of the letters of the word GUMTREE if:

a there are no restrictions,

b the Es are together,

c the Es are separated by:

i one,

ii two,

iii three,

iv four,

v five letters,

d the G is somewhere between the two Es,

e the M is somewhere to the left of both Es and the U is somewhere between them,

f the G is somewhere to the left of the U and the M is somewhere to the right of the U.

17 If the letters of the word GUMTREE and the letters of the word KOALA are combined and arranged into a single twelve-letter word, in how many of these arrangements do the letters of KOALA appear in their correct order, but not necessarily together?

18 Bob is about to hang his eight shirts in the wardrobe. He has four different styles of shirt, with two identical shirts in each style. How many different arrangements are possible if no two identical shirts are next to one another?

19 [Derangements and the number e]

A *derangement* was defined in question 22 of Exercise 14C as a permutation that leaves no letter unmoved, and $D(n)$ was defined as the number of derangements of n letters. Another approach to finding a formula for $D(n)$ uses the *inclusion–exclusion principle* introduced in the Enrichment section of Exercise 12C. Consider, for example, the derangements of ABCD.

a How many permutations are there of ABCD?

b How many of these leave in its original position:

i A,

ii B,

iii C,

iv D?

Each of these numbers will need to be subtracted from the answer to part **a**.

c In doing part **b**, however, we have subtracted some of the permutations twice. For example, some of them would leave both A and B unmoved. Thus we need to add back the number of permutations that leave any two particular letters unmoved. How many of these are there?

d Now you will need to subtract the number of permutations that leave three letters unmoved, and so on. Hence find an expression for $D(4)$.

e Rearrange your expression into the form $D(4) = 4!\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right)$.

f Explain how this can be generalised to

$$D(n) = n!\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right).$$

g We shall prove in an Enrichment question of the Year 12 volume that the sequence in the brackets of part **f** above converges to $\frac{1}{e}$ as $n \to \infty$. Give both a combinatorial and a probabilistic interpretation of this result in terms of the ratio of the numbers of permutations and derangements of n letters.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **16 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 8 and PP 9 are probability questions: they combine the counting of this lesson with $P(E) = \frac{|E|}{|S|}$ from the revision page. PP 10 and PP 11 count paths on a grid (Concept 2).

PP 1 **Knox Grammar School, 2022, Higher School Certificate Trial Examination, Mathematics Extension 1, Q7** 1 mark

In how many ways can the letters of the word EXCELLENT be arranged?

A. 3024 B. 30240

C. 60480 D. 362880

Hint: how many times does each letter of EXCELLENT appear?

PP 2 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q11(a)** 1 mark

Six chairs are to be painted. Two of the chairs are to be painted black.
The other four are to be painted blue, green, yellow and grey.
The chairs are then arranged in a row.
How many possible arrangements are there for the colours of the chairs?

Hint: once they are painted, which chairs look the same?

PP 3 **Sydney Technical High School, 2023, Extension 1 Mathematics, Trial HSC, Q1** 1 mark

The number of different arrangements of the letters of the word SPUDDLE which begins with the letter P is:

A. $\frac{7!}{2!}$

B. $\frac{6!}{2!}$

C. $\frac{6!}{5!\,2!}$

D. $\frac{7!}{5!\,2!}$

Hint: deal with the restriction first (Lesson 14B).

PP 4 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q12(a)** 1 mark

How many different ten-letter arrangements can be made using the letters of the word TABLECLOTH?

Hint: which letters of TABLECLOTH are repeated?

PP 5 **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q11(c)** 2 marks

Which boy's name has more unique arrangements of letters: MILO or SAVVAS? Show your working.

Hint: use Box 10 for each name.

PP 6 **Baulkham Hills High School, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q8** `1 mark`

In how many ways can the letters of the word **EXTENSION** be arranged if the letter E's are to be together?

A. $\dfrac{9!}{2!\,2!}$

B. $\dfrac{9!}{2!}$

C. $\dfrac{8!}{2!\,2!}$

D. $\dfrac{8!}{2!}$

Hint: think of the Es as one item.

PP 7 **St George Girls High School, 2020, Trial HSC Examination, Mathematics Extension 1, Q11(c)(i) (ii)** `3 marks`

Consider the word STATISTICS.

(i) How many arrangements of the letters are there? 1

(ii) How many arrangements of the letters are there where the A and C are next to each other? 2

Hint: start by counting how many times each letter of STATISTICS appears.

PP 8 **Normanhurst Boys High School, 2021, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q3** 1 mark

The letters in the word CALCULUS were randomly rearranged.
What is the probability that C is at the start and end of the new arrangement?

(A) $\frac{1}{6}$

(B) $\frac{1}{14}$

(C) $\frac{1}{28}$

(D) $\frac{1}{56}$

Hint: first count all the different arrangements of CALCULUS.

PP 9 **Ascham School, 2020, Trial HSC Examination, Mathematics Extension 1, Q2** 1 mark

Each of the letters A, A, C, H, M, and S is written on a separate card. The cards are drawn at random from a hat and placed next to each other to form a word. What is the probability that the word ASCHAM appears?

A. $\frac{1}{60}$

B. $\frac{1}{360}$

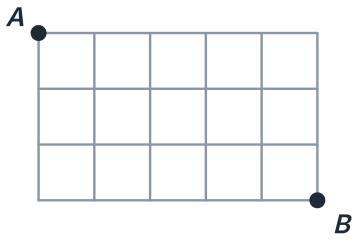
C. $\frac{1}{720}$

D. $\frac{1}{50}$

Hint: how many different words can the six cards make?

PP 10 **Normanhurst Boys High School, 2022, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(e)** 2 marks

The diagram shows a grid consisting of unit squares. Chris needs to travel from point A to point B but can only do so by moving right or down along the grid lines.

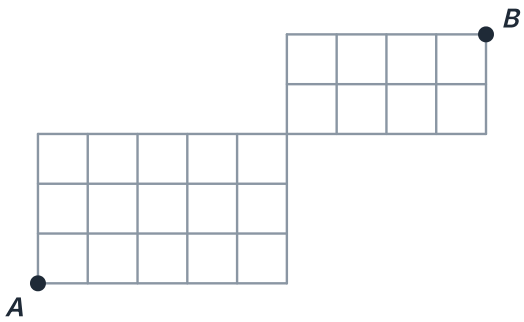


How many paths are there for Chris?

Hint: describe each path as a list of single steps.

PP 11 **St Aloysius' College, 2024, Trial Higher School Certificate Examination, Mathematics Extension 1, Q8** 1 mark

Moving only right or up, how many different paths exist to get from Point A to Point B?



(A) 45

(B) 840

(C) 480480

(D) $16!$

Hint: where do the two grids meet?

PP 12 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q10** 1 mark

Out of a group of 10 professional tiddlywinks players, 6 are to be selected for the final round of a competition. 4 of those 6 will be placed as 1st, 2nd, 3rd and 4th.
In how many ways can this process be carried out?

(A) $\dfrac{10!}{6!\,4!}$

(B) $\dfrac{10!}{4!\,4!}$

(C) $\dfrac{10!}{4!\,2!}$

(D) $\dfrac{10!}{6!}$

Hint: give each player a label that says what happens to them.

LESSON 14E

Counting unordered selections

This lesson turns from the ordered selections of the previous three lessons to the counting of unordered selections. An unordered selection of distinct objects chosen from a certain set is just a subset of the set. So given a set S with n members, there are two basic questions to ask: how many subsets does S have altogether, and how many r-member subsets does S have, for $r = 0, 1, \ldots, n$?

CONCEPT 1 · counting the subsets of a set

AN EXAMPLE OF THE RESULT

Let S be the five-member set $S = \{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}\}$. Here is the list of all the subsets of S, arranged by the number of members:

1	0-member subset:	$\varnothing$
5	1-member subsets:	{A}, {B}, {C}, {D}, {E}
10	2-member subsets:	{A, B}, {A, C}, {A, D}, {A, E}, {B, C}, {B, D}, {B, E}, {C, D}, {C, E}, {D, E}
10	3-member subsets:	{A, B, C}, {A, B, D}, {A, B, E}, {A, C, D}, {A, C, E}, {A, D, E}, {B, C, D}, {B, C, E}, {B, D, E}, {C, D, E}
5	4-member subsets:	{A, B, C, D}, {A, B, C, E}, {A, B, D, E}, {A, C, D, E}, {B, C, D, E}
1	5-member subset:	{A, B, C, D, E}

This makes $1 + 5 + 10 + 10 + 5 + 1 = 32$ subsets altogether. We will now establish the general results by finding clear reasons for these results.

HOW MANY SUBSETS DOES S HAVE ALTOGETHER?

Choosing a subset of S requires looking at each member of S in turn and deciding whether to include it in the subset or not. Let us write the code Y for 'Yes' if it is included, and N for 'No' if it is not included. Using this code, each subset of S corresponds to a five-letter word made up of Ys and Ns. For example,

YYNYN $\longleftrightarrow$ {A, B, D} and NNNNY $\longleftrightarrow$ {E}.

We can now use this code to count the total number of subsets. Thus we are counting the number of 5-letter words made up only of Ys and Ns:

$$\begin{aligned}\textbf{Total number of subsets} &= \textbf{number of 5-letter words made up of Ys and Ns}\\ &= 2^5\\ &= 32.\end{aligned}$$

HOW MANY 3-MEMBER SUBSETS DOES S HAVE?

We can also use the code to count the number of 3-member subsets, because that is the number of 5-letter words consisting of 3 Ys and 2 Ns (Box 11 of Lesson 14D):

$$\begin{aligned}\textbf{Number of 3-member subsets} &= \textbf{number of 5-letter words with 3 Ys and 2 Ns}\\ &= \frac{5!}{3! \times 2!}\\ &= 10.\end{aligned}$$

THE GENERAL FORMULAE AND THE NOTATION

These two results are easily generalised to any finite set S with n elements:

$$\textbf{total number of subsets} = 2^n$$

$$\textbf{number of } r\textbf{-member subsets} = \frac{n!}{r!\,(n-r)!}, \quad \text{for } r = 0, 1, \ldots, n.$$

The notation ${}^n\mathrm{C}_r$, or alternatively $\binom{n}{r}$, is used for the number of r-member subsets of S. Both are spoken as 'n choose r'. Summarising all this:

BOX 12 COUNTING THE SUBSETS OF A SET

Suppose that S is a finite set with n elements, and suppose that $0 \le r \le n$.

- Total number of subsets of $S = 2^n$.
- Number of r-member subsets of $S = {}^n\mathrm{C}_r = \dfrac{n!}{r!\,(n-r)!}$.
- ${}^n\mathrm{C}_0 + {}^n\mathrm{C}_1 + {}^n\mathrm{C}_2 + \cdots + {}^n\mathrm{C}_n = 2^n$.

The symbol ${}^n\mathrm{C}_r$ or $\binom{n}{r}$, both spoken as 'n choose r', denotes the number of r-subsets of an n-member set, where $0 \le r \le n$.

The last bullet point follows because LHS and RHS are both counting the total number of subsets of S.

The notations ${}^n\mathrm{P}_r$ and ${}^n\mathrm{C}_r$ are intended to be similar. The letter C stands for 'combination' (which is an old term for 'unordered selection'), just as the letter P stands for 'permutation'. By a convenient, but false, etymology, the letter C also stands for 'Choose', which is the origin of the more recent convention of saying 'n choose r' for ${}^n\mathrm{C}_r$.

Scientific calculators have a button labelled ${}^n\mathrm{C}_r$ which will find values of ${}^n\mathrm{C}_r$. For low values of n and r, the answers are exact, but for high values they are only approximations. As with ${}^n\mathrm{P}_r$, however, you should make a practice of evaluating these numbers by hand when it is reasonable to do so, because such calculations greatly help the intuition.

A PROOF MOVING FROM ORDERED SELECTIONS TO UNORDERED SELECTIONS

There is another way to prove the formula for ${}^{n}\mathrm{C}_{r}$. In Lesson 14B we saw that the number of three-letter words formed without repetition from the five letters A, B, C, D and E is ${}^{5}\mathrm{P}_{3} = 5 \times 4 \times 3 = 60$. When we turn to unordered selections, however, there are six distinct words which all correspond, for example, to the three-member subset {B, C, E}:

BCE, BEC, CBE, CEB, EBC, ECB $\longleftrightarrow$ {B, C, E}

The reason for this is that there are ${}^{3}\mathrm{P}_{3} = 3 \times 2 \times 1 = 6$ ways of ordering the subset {B, C, E}. Thus the correspondence between three-letter words and three-member subsets is many-to-one, with a six-fold overcounting. Hence the number of three-member subsets is $60 \div 6 = 10$, as required.

In general, ${}^{n}\mathrm{P}_{r} = \dfrac{n!}{(n-r)!}$ words of r letters can be formed without repetition from the members of an n-member set S. But every r-member subset can be ordered in ${}^{r}\mathrm{P}_{r} = r!$ ways, so the correspondence between the ordered selections and the unordered selections is many-to-one with overcounting by a factor of $r!$. Hence the number of (unordered) r-member subsets of the set S is

$$ {}^{n}\mathrm{P}_{r} \div {}^{r}\mathrm{P}_{r} = \frac{n!}{(n-r)!} \div r! = \frac{n!}{r! \times (n-r)!} = {}^{n}\mathrm{C}_{r}. $$

TUTOR DEMO **Counting subsets and calculating ${}^{n}\mathrm{C}_{r}$**

Let $T = \{1, 2, 3, 4, 5, 6\}$.

a How many subsets does T have altogether?

b How many 4-member subsets does T have?

Use the formula for ${}^{n}\mathrm{C}_{r}$ to evaluate:

c ${}^{9}\mathrm{C}_{4}$

d ${}^{n}\mathrm{C}_{4}$

e Find ${}^{15}\mathrm{C}_{4}$, leaving your answer factored into primes.

YOUR TURN **Same method**

Let $U = \{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}\}$.

a How many subsets does U have altogether?

b List the 2-member subsets of U, and check your list with the formula.

Use the formula for ${}^{n}\mathrm{C}_{r}$ to evaluate:

c ${}^{7}\mathrm{C}_{3}$

d ${}^{n}\mathrm{C}_{2}$

e Find ${}^{12}\mathrm{C}_{5}$, leaving your answer factored into primes.

WORKED EXAMPLE 15 **Keep for homework**

Use the formula for ${}^{n}\mathrm{C}_{r}$ to evaluate:

a ${}^{8}\mathrm{C}_{5}$

b ${}^{n}\mathrm{C}_{3}$

SOLUTION

a

$$\begin{aligned} {}^{8}\mathrm{C}_{5} &= \frac{8!}{3! \times 5!} \\ &= \frac{8 \times 7 \times 6 \times 5!}{3 \times 2 \times 1 \times 5!} \\ &= 56 \end{aligned}$$

b

$$\begin{aligned} {}^{n}\mathrm{C}_{3} &= \frac{n(n-1)(n-2)(n-3)!}{(n-3)! \times 3!} \\ &= \frac{n(n-1)(n-2)}{6} \end{aligned}$$

WORKED EXAMPLE 16 **Keep for homework**

Find ${}^{16}\mathrm{C}_{5}$, leaving your answer factored into primes.

SOLUTION

$$\begin{aligned} {}^{16}\mathrm{C}_{5} &= \frac{16 \times 15 \times 14 \times 13 \times 12}{1 \times 2 \times 3 \times 4 \times 5} \\ &= 2 \times 14 \times 13 \times 12 \\ &= 2^{4} \times 3 \times 7 \times 13 \quad (\text{Check this on the calculator.}) \end{aligned}$$

CONCEPT 2 · a natural (or canonical) correspondence

A NATURAL (OR CANONICAL) CORRESPONDENCE: ${}^nC_r = {}^nC_{n-r}$

Suppose that two people are to be chosen from five to make afternoon tea. This task can be done in two ways:

- Choose the two people out of five who will *make the tea*.
- Choose the three people out of five who will *not make the tea*.

Thus for every choice of two people out of five, the remaining three people is a corresponding choice of three people out of five. This confirms that ${}^5C_2 = {}^5C_3$.

But it also gives a one-to-one correspondence between the two-member subsets and the three-member subsets of a five-member set:

{A, B} ⟷ {C, D, E}	{B, D} ⟷ {A, C, E}
{A, C} ⟷ {B, D, E}	{B, E} ⟷ {A, C, D}
{A, D} ⟷ {B, C, E}	{C, D} ⟷ {A, B, E}
{A, E} ⟷ {B, C, D}	{C, E} ⟷ {A, B, D}
{B, C} ⟷ {A, D, E}	{D, E} ⟷ {A, B, C}

In this *natural* or *canonical correspondence* between the 2-member subsets and the 3-member subsets, every subset T is paired with its complement $\overline{T}$:

$$T \longleftrightarrow \overline{T}$$

The situation is easily generalised:

BOX 13 A CANONICAL CORRESPONDENCE

Let n and r be whole numbers with $0 \leq r \leq n$, and let S be an n-member set.

- ${}^nC_r = {}^nC_{n-r}$
- The r-member subsets of S are *naturally* or *canonically* paired up with the $(n-r)$-subsets of S by pairing each subset with its complement.

The correspondence is by no means restricted to mathematics. In normal language, a situation can often be described just as well by saying what it is not as by saying what it is. We are all familiar with ‘days when no rain fell’ or ‘unforgivable actions’ or ‘invisible enemies’.

TUTOR DEMO **Choose the ones that are left out**

a Write ${}^{9}\mathrm{C}_{2}$ and ${}^{9}\mathrm{C}_{7}$ in factorial notation, showing that they are equal.
b Use the canonical correspondence to evaluate ${}^{20}\mathrm{C}_{18}$.
c In how many ways can seven people out of nine be chosen to make afternoon tea?

YOUR TURN **Same method**

a Write ${}^{8}\mathrm{C}_{3}$ and ${}^{8}\mathrm{C}_{5}$ in factorial notation, showing that they are equal.
b Use the canonical correspondence to evaluate ${}^{15}\mathrm{C}_{13}$.
c In how many ways can five people out of seven be chosen to make afternoon tea?

WORKED EXAMPLE 17 **Keep for homework**

Write ${}^{7}\mathrm{C}_{2}$ and ${}^{7}\mathrm{C}_{5}$ in factorial notation, showing that they are equal.

SOLUTION

$${}^{7}\mathrm{C}_{2} = \frac{7!}{2! \times 5!} \qquad {}^{7}\mathrm{C}_{5} = \frac{7!}{5! \times 2!}$$

CONCEPT 3 · using combinations in counting problems

USING ${}^nC_r$ IN COUNTING PROBLEMS

Here are a number of examples of counting problems. A committee, a team or a hand of cards is an unordered selection, so it is counted with ${}^nC_r$. When the selection is made from two separate groups, such as women and men, choose from each group separately and multiply.

WARNING WORDS

As always, words such as ‘at least’, ‘at most’, ‘not’ and ‘excluding’ should always be regarded as warnings that the problem may best be solved by considering the complementary situation.

TUTOR DEMO **Include, exclude, and choose from two groups**

Twelve people meet to play doubles tennis.
a In how many ways can four people be selected from this group to play the first game? (Ignore the subsequent organisation into pairs.)
b How many of these ways will include Maria and exclude Alex?
c If there are five women and seven men, in how many ways can two men and two women be chosen for this game?
d Again with five women and seven men, in how many ways will women be in the majority?

YOUR TURN **Same method**

Nine people meet to play doubles tennis.
a In how many ways can four people be selected from this group to play the first game? (Ignore the subsequent organisation into pairs.)
b How many of these ways will include Maria and exclude Alex?
c If there are four women and five men, in how many ways can two men and two women be chosen for this game?
d Again with four women and five men, in how many ways will women be in the majority?

WORKED EXAMPLE 18 **Keep for homework**

Ten people meet to play doubles tennis.
a In how many ways can four people be selected from this group to play the first game? (Ignore the subsequent organisation into pairs.)
b How many of these ways will include Maria and exclude Alex?
c If there are four women and six men, in how many ways can two men and two women be chosen for this game?
d Again with four women and six men, in how many ways will women be in the majority?

SOLUTION

a $\text{Number of ways} = {}^{10}\text{C}_4 = 210.$

b Because Maria is included, three further people must be chosen, and because Alex is excluded, there are now eight people to choose these three from.

Hence $\text{number of ways} = {}^{8}\text{C}_3 = 56.$

c

$$\begin{aligned}\text{Number of ways of choosing the women} &= {}^{4}\text{C}_2 = 6,\\ \text{number of ways of choosing the men} &= {}^{6}\text{C}_2 = 15,\\ \text{so number of ways of choosing all four} &= 15 \times 6 = 90.\end{aligned}$$

d

$$\begin{aligned}\text{Number of ways with one man and three women} &= {}^{6}\text{C}_1 \times {}^{4}\text{C}_3 = 24,\\ \text{number of ways with four women} &= 1,\\ \text{so number of ways with a majority of women} &= 24 + 1 = 25.\end{aligned}$$

CONCEPT 4 · subsets with conditions

SUBSETS WITH CONDITIONS

'At least two numbers' is a warning word. Count the subsets that are *not* wanted (the 0-member and 1-member subsets) and subtract them from the total 2^n of Box 12. A condition such as 'does not contain 8' removes 8 from the set before counting. A condition such as 'does contain 10' means that 10 has already been chosen, and only the rest of the subset remains to be chosen.

TUTOR DEMO **Subtract the unwanted subsets**

Let $S = \{3, 6, 9, 12, 15, 18, 21\}$ be the set consisting of the first seven positive multiples of 3.

a How many subsets of S contain at least two numbers?

b How many subsets with at least two numbers do not contain 9?

c How many subsets with at least two numbers do not contain 9 but do contain 21?

YOUR TURN **Same method**

Let $S = \{5, 10, 15, 20, 25\}$ be the set consisting of the first five positive multiples of 5.

a How many subsets of S contain at least two numbers?

b How many subsets with at least two numbers do not contain 10?

c How many subsets with at least two numbers do not contain 10 but do contain 25?

WORKED EXAMPLE 19 **Keep for homework**

Let $S = \{2, 4, 6, 8, 10, 12\}$ be the set consisting of the first six positive even numbers.
a How many subsets of S contain at least two numbers?
b How many subsets with at least two numbers do not contain 8?
c How many subsets with at least two numbers do not contain 8 but do contain 10?

SOLUTION

a

$$\text{Number of 1-member and 0-member subsets} = {}^6\text{C}_1 + {}^6\text{C}_0 = 7,$$
$$\text{so number with at least 2 members} = 2^6 - 7 = 57.$$

b We consider now the 5-member set $T = \{2, 4, 6, 10, 12\}$.

$$\text{Number of 1-member and 0-member subsets} = {}^5\text{C}_1 + {}^5\text{C}_0 = 6,$$
$$\text{so number with at least 2 members} = 2^5 - 6 = 26.$$

c Because 10 has already been chosen, we need to choose subsets with at least one member from the four-member set $U = \{2, 4, 6, 12\}$.

$$\text{Number of 0-member subsets} = 1 \quad (\text{the empty set}),$$
$$\text{so number of such subsets} = 2^4 - 1 = 15.$$

CONCEPT 5 · a harder question: points and triangles

CHOOSING POINTS AS VERTICES

A triangle is fixed by choosing its three vertices, and the order of choosing them does not matter. So triangles drawn with given points as vertices are counted with ${}^n\text{C}_r$. To form two triangles with distinct vertices, first choose all six vertices, and then decide which of them go together.

TUTOR DEMO **Two triangles from points on a circle**

Eight points $A_1, A_2, \ldots, A_8$ are arranged in order around a circle.
a How many triangles can be drawn with these points as vertices?
b How many pairs of such triangles can be drawn, if the vertices of the two triangles are distinct?
c In how many such pairs will the triangles:
i not overlap,
ii overlap?

YOUR TURN **Same method**

Seven points $A_1, A_2, \ldots, A_7$ are arranged in order around a circle.
a How many triangles can be drawn with these points as vertices?
b How many pairs of such triangles can be drawn, if the vertices of the two triangles are distinct?
c In how many such pairs will the triangles:
i not overlap,
ii overlap?

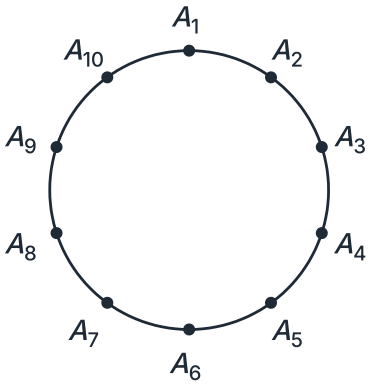
WORKED EXAMPLE 20 **Keep for homework**

[A harder question]
Ten points $A_1, A_2, \ldots, A_{10}$ are arranged in order around a circle.
a How many triangles can be drawn with these points as vertices?
b How many pairs of such triangles can be drawn, if the vertices of the two triangles are distinct?
c In how many such pairs will the triangles:
i not overlap,
ii overlap?

SOLUTION

a To form a triangle, we must choose 3 points out of 10, so

$$\text{number of triangles} = {}^{10}\text{C}_3 = 120.$$



b To form two triangles, first choose 6 points out of 10, which can be done in ${}^{10}\text{C}_6 = 210$ ways.

Take any one of those 6 points, then choose the other 2 points in its triangle, which can be done in ${}^{5}\text{C}_2 = 10$ ways.

The second triangle is then drawn using the other 3 points.

$$\text{Hence number of pairs of triangles} = 210 \times 10 = 2100.$$

c To form two non-overlapping triangles, we first choose 6 points out of 10, which again can be done in ${}^{10}\text{C}_6 = 210$ ways. These 6 points can be made into two non-overlapping triangles in 3 ways, by arranging the 6 points in cyclic order, and choosing 3 adjacent points.

i Hence, $\text{the number of non-overlapping pairs} = 210 \times 3 = 630.$

ii By subtraction, $\text{number of overlapping pairs} = 2100 - 630 = 1470.$

TEXTBOOK HOMEWORK · Exercise 14E

FOUNDATION

1 Two people are chosen from a group of five people called P, Q, R, S and T. List all possible combinations, and find how many there are.

2 Find in how many ways you can form a group of:

a two people from a choice of seven,

b three people from a choice of seven,

c two people from a choice of six,

d five people from a choice of nine.

3 **a** Find how many possible combinations there are if, from a group of ten people:

i two people are chosen,

ii eight people are chosen.

b Why are the answers identical?

4 From a party of twelve men and eight women, find how many groups can be chosen consisting of:

a five men and three women,

b four women and four men.

5 Four numbers are to be selected from the set of the first eight positive integers. Find how many possible combinations there are if:

a there are no restrictions,

b there are two odd numbers and two even numbers,

c there is exactly one odd number,

d all the numbers must be even,

e there is at least one odd number.

7 A committee of five is to be chosen from six men and eight women. Find how many committees are possible if:

a there are no restrictions,

b all members are to be female,

c all members are to be male,

d there are exactly two men,

e there are four women and one man,

f there is a majority of women,

g a particular man must be included,

h a particular man must not be included.

DEVELOPMENT

8 **a** What is the number of combinations of the letters of the word EQUATION taken four at a time (without repetition)?

b How many of the four-letter combinations contain four vowels?

c How many of the four-letter combinations contain the letter Q?

9 A team of seven netballers is to be chosen from a squad of twelve players A, B, C, D, E, F, G, H, I, J, K and L. In how many ways can they be chosen:

a with no restrictions,

b if the captain C is to be included,

c if J and K are both to be excluded,

d if A is included but H is not,

e if one of F and L is to be included and the other excluded?

10 **a** Consider the digits 9, 8, 7, 6, 5, 4, 3, 2, 1 and 0. Find how many five-digit numbers are possible if the digits are to be in:

i descending order,

ii ascending order.

b Why do these two questions involve unordered selections?

11 Twelve people arrive at a restaurant. There is one table for six, one table for four and one table for two. In how many ways can they be assigned to a table?

12 Twenty students, ten male and ten female, are to travel from school to the sports ground. Eight of them go in a minibus, six of them in cars, four of them on bikes and two walk.

a In how many ways can they be distributed for the trip?

b In how many ways can they be distributed if none of the boys walk?

13 Ten points $P_1, P_2, \ldots, P_{10}$ are chosen in a plane, no three of the points being collinear.

a How many lines can be drawn through pairs of the points?

b How many triangles can be drawn using the given points as vertices?

c How many of these triangles have P_1 as one of their vertices?

d How many of these triangles have P_1 and P_2 as vertices?

14 Ten points are chosen in a plane. Five of the points are collinear, but no other set of three of the points is collinear.

a How many sets of three points can be selected from the five that are collinear?

b How many triangles can be formed using three of the ten points as vertices?

15 From a standard deck of 52 playing cards, find how many five-card hands can be dealt:

a consisting of black cards only,

b consisting of diamonds only,

c containing all four kings,

d consisting of three diamonds and two clubs,

e consisting of three twos and another pair,

f consisting of one pair and three of a kind.

16 **a** In how many ways can a group of six people be divided into:

i two unequal groups (neither group being empty),

ii two equal groups?

b Repeat part **a** for four people.

c Repeat part **a** for eight people.

17 Find how many diagonals there are in:

a a quadrilateral,

b a pentagon,

c a decagon,

d a polygon with n sides.

18 Twelve points are arranged in order around a circle.

a How many triangles can be drawn with these points as vertices?

b In how many pairs of such triangles are the vertices of the two triangles distinct?

c In how many such pairs will the triangles:

i not overlap,

ii overlap?

19 Let $S = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$ be the set of the first ten positive odd integers.

a How many subsets does S have?

b How many subsets of S contain at least three numbers?

c How many subsets with at least three numbers do not contain 7?

d How many subsets with at least three numbers do not contain 7 but do contain 19?

20 In how many ways can two numbers be selected from $1, 2, \ldots, 8, 9$ so that their sum is:

a even,

b odd,

c divisible by 3,

d divisible by 5,

e divisible by 6?

21 There are ten basketballers in a team. Find in how many ways:

a the starting five can be chosen,

b they can be split into two teams of five.

22 Nine players are to be divided into two teams of four and one umpire.

a In how many ways can the teams be formed?

b If two particular people cannot be on the same team, how many different combinations are possible?

23 By considering their prime factorisations, find the number of positive divisors of:

a $2^3 \times 3^2$

b $1\,000\,000$

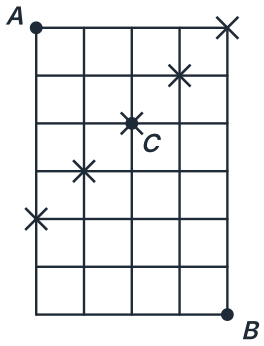
c $315\,000$

d $2^a \times 5^b \times 13^c$

24 **a** The six faces of a number of identical cubes are painted in six distinct colours. How many different cubes can be formed?

b A die fits perfectly into a cubical box. How many ways are there of putting the die into the box?

25 The diagram shows a 6×4 grid. The aim is to walk from the point A in the top left-hand corner to the point B in the bottom right-hand corner by walking along the black lines either downwards or to the right. A single move is defined as walking along one side of a single small square, thus it takes you ten moves to get from A to B.



a Find how many different routes are possible:

i without restriction,

ii if you must pass through C,

iii if you cannot move along the top line of the grid,

iv if you cannot move along the second row from the top of the grid.

b Notice that every route must pass through one and only one of the five crossed points. Hence prove that

$$^{10}\mathrm{C}_4 = {}^{4}\mathrm{C}_0 \times {}^{6}\mathrm{C}_4 + {}^{4}\mathrm{C}_1 \times {}^{6}\mathrm{C}_3 + {}^{4}\mathrm{C}_2 \times {}^{6}\mathrm{C}_2 + {}^{4}\mathrm{C}_3 \times {}^{6}\mathrm{C}_1 + {}^{4}\mathrm{C}_4 \times {}^{6}\mathrm{C}_0.$$

c Draw another suitable diagonal and, using a method similar to that in part **b**, prove that

$$^{10}\mathrm{C}_4 = {}^{5}\mathrm{C}_0 \times {}^{5}\mathrm{C}_4 + {}^{5}\mathrm{C}_1 \times {}^{5}\mathrm{C}_3 + {}^{5}\mathrm{C}_2 \times {}^{5}\mathrm{C}_2 + {}^{5}\mathrm{C}_3 \times {}^{5}\mathrm{C}_1 + {}^{5}\mathrm{C}_4 \times {}^{5}\mathrm{C}_0.$$

d Draw up a similar 6×6 grid, then using the same idea as that used in parts **b** and **c**, prove that

$$^{12}\mathrm{C}_6 = \left({}^{6}\mathrm{C}_0\right)^2 + \left({}^{6}\mathrm{C}_1\right)^2 + \left({}^{6}\mathrm{C}_2\right)^2 + \left({}^{6}\mathrm{C}_3\right)^2 + \left({}^{6}\mathrm{C}_4\right)^2 + \left({}^{6}\mathrm{C}_5\right)^2 + \left({}^{6}\mathrm{C}_6\right)^2.$$

ENRICHMENT

26 A piece of art receives an integer mark from zero to 100 for each of the categories design, technique and originality. In how many ways is it possible to score a total mark of 200?

27 How many different combinations are there of three different integers between one and thirty inclusive such that their sum is divisible by three?

28 **a** How many doubles tennis games are possible, given a group of four players?

b In how many ways can two games of doubles tennis be arranged, given a group of eight players?

c Six men and six women are to play in three games of doubles tennis. Find how many ways the pairings can be arranged if:

i there are no restrictions,

ii each game is to be a game of mixed doubles.

29 Referring to Question 18 of Exercise 14D, Bob's interest in shirts has matured recently: he now has $2n$ shirts, with two identical shirts in each of n distinct styles. He still wants to hang his shirts in the wardrobe so that no two identical shirts are next to one another.

a Show that the number of allowed arrangements is
$${}^{n}\mathrm{C}_0\,(2n)!\,2^{-n} - {}^{n}\mathrm{C}_1\,(2n-1)!\,2^{1-n} + {}^{n}\mathrm{C}_2\,(2n-2)!\,2^{2-n} + \cdots + (-1)^n\,{}^{n}\mathrm{C}_n\,n!\,2^0.$$

b Evaluate this number for $n = 1, 2, 3, 4, 5$ and 6, and find the corresponding ratio of the total number of arrangements to the number of allowed arrangements.

30 Computing is based on *binary strings*, which are sequences of 0s and 1s, such as 00101 and 1001110. The problem in this question is to find the number N of binary strings that contain exactly a 0s and at most b 1s, where a and b are whole numbers. We will count such strings in two different ways, and hence prove an interesting identity.

a Using ${}^{n}\mathrm{C}_{r}$ notation, find how many binary strings with exactly a 0s contain:

i no 1s

ii one 1

iii two 1s

iv three 1s

v r 1s.

b Hence prove that $N = {}^{a}\mathrm{C}_{0} + {}^{a+1}\mathrm{C}_{1} + {}^{a+2}\mathrm{C}_{2} + \cdots + {}^{a+b}\mathrm{C}_{b}$.

c Let S be a binary string consisting of a 0s and r 1s, where r is at most b. Extend S to a longer binary string $S01\cdots1$ by adding 0 on the right, and then adding $b - r$ 1s. How many 0s and 1s are there in the resulting string, and what is its total length?

d Describe the inverse process by which a string of $(a+1)$ 0s and b 1s can be converted to a string of a 0s and at most b 1s, and show that the two processes are one-to-one correspondences.

e Hence prove that $N = {}^{a+b+1}\mathrm{C}_{b}$.

f Show that ${}^{a}\mathrm{C}_{0} + {}^{a+1}\mathrm{C}_{1} + {}^{a+2}\mathrm{C}_{2} + \cdots + {}^{a+b}\mathrm{C}_{b} = {}^{a+b+1}\mathrm{C}_{b}$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q6** 1 mark

How many different ways can 4 people be chosen from a group of 20 people?

A. 4 845
B. 160 000
C. 116 280
D. 240 000

Hint: does the order in which the 4 people are chosen matter?

PP 2 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q6** 1 mark

A school mathematics department consists of 5 male and 5 female teachers. How many different exam-writing committees comprising three teachers could be formed if there must be at least one teacher of each sex on the committee?

(A) 50
(B) 100
(C) 120
(D) 600

Hint: 'at least' is one of the warning words of Concept 3.

PP 3 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q3** 1 mark

A sporting team needs 5 members, which must include at least two students from Year 11 and at least two students from Year 12.
There are ten Year 11 students and fifteen Year 12 students available for selection.
In how many distinct ways can the team be chosen?

A. 10
B. 5775
C. 33 075
D. 396 900

Hint: what are the possible numbers of Year 11 students in the team?

PP 4 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q6** 1 mark

A team of r student representatives are chosen from class of n students where $r < n$.
This team should include a captain and a vice captain.
How many ways can this team be selected?

A. $n(n-1)\binom{n}{r}$
B. $2! \times \binom{n}{r}$
C. $r(r-1)\binom{n-2}{r}$
D. $n(n-1)\binom{n-2}{r-2}$

Hint: two members of the team have special roles. Which choices are ordered, and which are not?

PP 5 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q12(d)(i)(ii)** 3 marks

In a class of 27 students, there are 14 boys and 13 girls.
The class needs to elect two boys and two girls for the student council.

(i) In how many ways could the representatives be chosen? 2

(ii) If the class elected Henry and Olivia for the student council, in how many ways could the representatives now be chosen? 1

Hint: think about the boys and the girls separately.

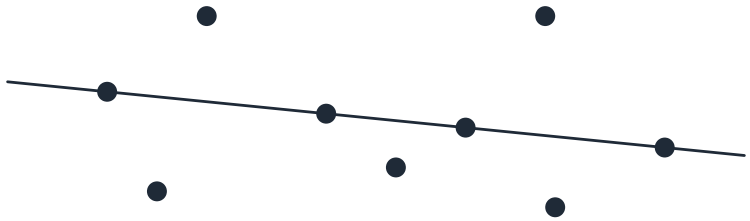
PP 6 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q11(e)** 2 marks

Peter and May are two of ten candidates for a committee.
How many ways can a committee of five be chosen, if Peter refuses to be on the same committee as May?

Hint: which committees break Peter's condition?

PP 7 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q13(a)** 2 marks

Nine points are drawn on a page as shown, with four of them lying on one line.
Other than these four points, no other set of three points is collinear.



How many different triangles can be drawn using the nine points as vertices?

Hint: which sets of three points do not make a triangle?

PP 8 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(e)(i)(ii)** 3 marks

A team of 12 basketball players is selected from a pool of 8 left-handed students and 16 right-handed students. How many ways can the team be chosen if:

(i) There are no restrictions? 1

(ii) The team has more left-handed students than right-handed students? 2

Hint: in (ii), what does 'more left-handed than right-handed' mean for a team of 12?

PP 9 **Normanhurst Boys High School, 2023, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(e)(i)(ii)** 2 marks

There are twelve people that need to be divided into groups.

(i) In how many ways can they be divided into two groups of six people? 1

(ii) In how many ways can they be divided into two groups of six people if two particular people must be in the same group? 1

Hint: the two groups have no names. Does it matter which group is chosen first?

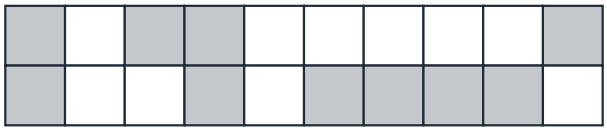
PP 10 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(d)** 2 marks

There are n dancers in a performing arts academy.
The trainer is to select a group of two or three of these dancers to perform in a certain concert.
The number of different groups of three dancers he can select is 3 times the number of different groups of two dancers.
Find n the number dancers in the performing arts academy.

Hint: Worked example 15 (b) writes ${}^{n}\mathrm{C}_{3}$ in terms of n.

PP 11 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q12(c)** `2 marks`

You are given two rows of square tiles, 10 tiles in each row. You are asked to shade 4 tiles in the top row and 6 tiles in the bottom row so that exactly two shaded tiles in the top row are directly above a shaded tile in the bottom row, An example of one such arrangement is shown.



In how many ways is this possible?

Hint: which choices have to be made, and in which row?

LESSON 14F

Using counting in probability

The purpose of this lesson is to apply the counting procedures of the last four lessons to questions about probability. In these more complicated questions, counting procedures are required for counting both the sample space and the event space. Then $P(E) = \dfrac{|E|}{|S|}$, as on the revision page.

CONCEPT 1 · counting the sample space and the event space

COUNTING THE SAMPLE SPACE AND THE EVENT SPACE

As always in this topic, the two questions that need to be asked are:

- Are the selections we are counting ordered or unordered?
- If they are ordered, is repetition allowed or not?

If the questions can be done with ordered selections or with unordered selections, it is usually easier to use unordered selections because the numbers are smaller.

USE THE SAME KIND OF SELECTION ON THE TOP AND THE BOTTOM

The sample space and the event space must be counted in the same way. If the sample space is the set of unordered hands, count the favourable unordered hands. If it is the set of ordered hands, count the favourable ordered hands.

TUTOR DEMO **Any order, or that order**

Four cards are dealt from a pack of 52.
a Find the probability that one heart and three spades are dealt, in any order.
b Find the probability that one heart and three spades are dealt in the order ♡♠♠♠.

YOUR TURN **Same method**

Two cards are dealt from a pack of 52.
a Find the probability that one red card and one black card are dealt, in any order.
b Find the probability that a red card and then a black card are dealt, in that order.

WORKED EXAMPLE 21 **Keep for homework**

Three cards are dealt from a pack of 52.
a Find the probability that one club and two hearts are dealt, in any order.
b Find the probability that one club and two hearts are dealt in that order.

SOLUTION

a Let the sample space be the set of all unordered selections of 3 cards from 52,

$$\text{so} \qquad \text{number of unordered hands} = {}^{52}\mathrm{C}_3 = \frac{52 \times 51 \times 50}{3 \times 2 \times 1}.$$

We can now choose the hand by choosing 1 club from 13 in ${}^{13}\mathrm{C}_1 = 13$ ways,
and choosing the 2 hearts from 13 in ${}^{13}\mathrm{C}_2 = 78$ ways,
so the hand can be chosen in 13×78 ways.

$$\text{Hence} \qquad P(\text{1 club and 2 hearts}) = 13 \times 78 \times \frac{3 \times 2 \times 1}{52 \times 51 \times 50} = \frac{39}{850}.$$

b Let the sample space be the set of all ordered selections of 3 cards from 52,

$$\text{so} \qquad \text{number of ordered hands} = {}^{52}\mathrm{P}_3 = 52 \times 51 \times 50,$$

$$\text{and} \qquad \text{number of such hands in the order } \clubsuit\heartsuit\heartsuit = 13 \times 13 \times 12.$$

$$\text{Hence} \qquad P(\clubsuit\heartsuit\heartsuit) = \frac{13 \times 13 \times 12}{52 \times 51 \times 50} = \frac{13}{850}.$$

Note: The answer to part **b** must be $\frac{1}{3}$ of the answer to part **a**, because in a hand with one club and two hearts, the club can be any one of three positions. This indicates that it would be quite reasonable to do part **a** using ordered selections, and to do part **b** using unordered selections, although the methods chosen above are more natural to the way in which each question was worded.

CONCEPT 2 · problems requiring a variety of methods

PROBLEMS REQUIRING A VARIETY OF METHODS

The sample spaces in the next two examples are easily found, but a variety of methods is needed to establish the sizes of the various event spaces.

- The first digit of a number cannot be zero. Deal with this difficulty first (Lesson 14B).
- 'The digits are distinct' means counting without replacement.
- Every unordered selection of distinct digits can be arranged in exactly one way with the digits in increasing order. So numbers with distinct digits in increasing order are counted by ${}^{n}\mathrm{C}_{r}$ (Lesson 14E).

TUTOR DEMO A four-digit number chosen at random

A four-digit number is chosen at random. Find the probability:
a that it is at least 7000,
b that it consists only of odd digits,
c that the digits are distinct,
d that the digits are distinct and in increasing order.

YOUR TURN Same method

A three-digit number is chosen at random. Find the probability:
a that it is at least 400,
b that it consists only of even digits,
c that the digits are distinct,
d that the digits are distinct and in increasing order.

WORKED EXAMPLE 22 **Keep for homework**

A five-digit number is chosen at random. Find the probability:
a that it is at least $60\,000$,
b that it consists only of even digits,
c that the digits are distinct,
d that the digits are distinct and in increasing order.

SOLUTION

The first digit of a five-digit number cannot be zero, giving nine choices, but the other digits can be any one of the ten digits.
Hence the number of five-digit numbers $= 9 \times 10 \times 10 \times 10 \times 10 = 90\,000$.

a To be at least $60\,000$, the first digit can be 6, 7, 8 or 9,
so the number of favourable numbers is $4 \times 10 \times 10 \times 10 \times 10 = 40\,000$.
Hence $P(\text{at least } 60\,000) = \frac{4}{9}$.

b If all the digits are even, there are four choices for the first digit (it cannot be zero) and five choices for each of the other four.
Hence number of such numbers $= 4 \times 5 \times 5 \times 5 \times 5 = 2500$,
and $P(\text{all digits are even}) = \dfrac{2500}{90\,000} = \dfrac{1}{36}$.

c This is counting without replacement:

1st digit	2nd digit	3rd digit	4th digit	5th digit
9	9	8	7	6

so number of such numbers $= 9 \times 9 \times 8 \times 7 \times 6$
and $P(\text{digits are distinct}) = \dfrac{9 \times 9 \times 8 \times 7 \times 6}{90\,000} = \dfrac{189}{625}$.

d Every *unordered* five-member subset of the set of nine non-zero digits can be arranged in exactly one way into a five-digit number with the digits in increasing order. (Note that the digit zero cannot be used, because a number can't begin with the digit zero.)
Hence number of such numbers = number of unordered 5-member subsets of $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
$= {}^9\mathrm{C}_5 = 126$,
so $P(\text{digits are distinct and in increasing order}) = \dfrac{126}{90\,000} = \dfrac{7}{5000}$.

CONCEPT 3 · a harder question: complements, the addition theorem and cases

THREE MORE METHODS FOR THE EVENT SPACE

Worked example 23 needs three more methods for its event spaces:

- **the complementary event:** 'at least one 4' is everything except 'no 4',
- **the addition theorem** $|A \cup B| = |A| + |B| - |A \cap B|$, for 'no 4 or no 5',
- **cases:** count the strings of digits first, then subtract the strings that begin with zero.

TUTOR DEMO **Four-digit numbers again**

A four-digit number is chosen at random. Find the probability that:
a the number contains at least one three,
b the number contains at least one three and at least one six,
c the number contains exactly two sevens,
d the number contains at least two sevens.

YOUR TURN **Same method**

A three-digit number is chosen at random. Find the probability that:
a the number contains at least one five,
b the number contains at least one five and at least one six,
c the number contains exactly two twos,
d the number contains at least two twos.

WORKED EXAMPLE 23 **Keep for homework**

[A harder question] Continuing with Worked example 22, find the probability that:
a the number contains at least one four,
b the number contains at least one four and at least one five,
c the number contains exactly three sevens,
d the number contains at least three sevens.

SOLUTION

a This can be approached using the complementary event:
number of five-digit numbers without a 4 $= 8 \times 9 \times 9 \times 9 \times 9 = 52\,488$,
so number of five-digit numbers with a 4 $= 90\,000 - 52\,488 = 37\,512$.
Hence $P(\text{at least one 4}) = \dfrac{37\,512}{90\,000} = \dfrac{521}{1250}$.

b This can be approached using the addition theorem:
number without a 4 $= 52\,488$,
similarly number without a 5 $= 52\,488$,
and number with no 5 and no 4 $= 7 \times 8 \times 8 \times 8 \times 8 = 28\,672$.
Hence number with no 5 or no 4 $= 52\,488 + 52\,488 - 28\,672 = 76\,304$,
and number with at least one 5 and at least one 4 $= 90\,000 - 76\,304 = 13\,696$.
Hence $P(\text{at least one 4 and at least one 5}) = \dfrac{13\,696}{90\,000} = \dfrac{856}{5625}$.

c Counting the number of five-digit numbers with exactly three 7s requires cases. First we count the five-digit strings with exactly three 7s, by first placing the three 7s and then choosing the first and second non-7 digits:

position of the three 7s	choose first non-7	choose second non-7
${}^5\mathrm{C}_3 = 10$	9	9

giving 810 such strings. Secondly, we must subtract the number of five-digit strings with exactly three 7s and beginning with zero:

position of the three 7s	choose the other non-7
${}^4\mathrm{C}_3 = 4$	9

giving 36 such strings. Hence there are $810 - 36 = 774$ such numbers, and
$P(\text{number has exactly three 7s}) = \dfrac{774}{90\,000} = \dfrac{43}{5000}$.

d The number $77\,777$ is the only five-digit number with five 7s. Any five-digit number with exactly four 7s has one of the five forms $*7777,\ 7*777,\ 77*77,\ 777*7,\ 7777*$, where the $*$ in $*7777$ is a non-zero digit. There are eight numbers of the first form, and nine of the other four forms, giving 44 numbers altogether.
Hence the number with at least three 7s $= 774 + 44 + 1 = 819$,
and $P(\text{at least three 7s}) = \dfrac{819}{90\,000} = \dfrac{91}{10\,000}$.

TEXTBOOK HOMEWORK · Exercise 14F

FOUNDATION

1 A committee of three is to be selected from the nine members in a club.

a How many different committees can be formed?

b If there are five men in the club, what is the probability that the selected committee consists entirely of males?

2 The integers from 1 to 10 inclusive are written on ten separate pieces of paper. Four pieces of paper are drawn at random. Find the probability that:

a the four numbers drawn are 1, 2, 3 and 6,

b the number 9 is one of the numbers drawn,

c the number 8 is not drawn,

d the number 7 is drawn but the number 1 is not.

3 A bag contains three red, seven yellow and five blue balls. If three balls are drawn from the bag simultaneously, find the probability that:

a all three balls are yellow,

b all the balls are of the same colour,

c there are two red balls and one blue ball,

d all the balls are of different colours.

4 A sports committee of five members is to be chosen from eight AFL footballers and seven soccer players. Find the probability that the committee will contain:

a only AFL footballers,

b only soccer players,

c three soccer players and two AFL footballers,

d at least one soccer player,

e at most one soccer player,

f Ian, a particular soccer player.

5 From a standard pack of 52 cards, three are selected at random. Find the probability that:

a they are the jack of spades, the two of clubs and the seven of diamonds,

b all three are aces,

c they are all diamonds,

d they are all of the same suit,

e they are all picture cards,

f two are red and one is black,

g one is a seven, one is an eight and one is a nine,

h two are sevens and one is a six,

i exactly one is a diamond,

j at least two of them are diamonds.

6 Repeat the previous question if the cards are selected from the pack one at a time, and each card is replaced before the next one is drawn.

7 Three boys and three girls are to sit in a row. Find the probability that:

a the boys and girls alternate,

b the boys sit together and the girls sit together,

c two specific girls sit next to one another.

8 A family of five are seated in a row at the cinema. Find the probability that:

a the parents sit on the end and the three children are in the middle,

b the parents sit next to one another.

9 Six people, of whom Patrick and Jessica are two, arrange themselves in a row. Find the probability that:

a Patrick and Jessica occupy the end positions,

b Patrick and Jessica are not next to each other.

DEVELOPMENT

10 The letters of PROMISE are arranged randomly in a row. Find the probability that:

a the word starts with R and ends with S,

b the letters P and R are next to one another,

c the letters P and R are separated by at least three letters,

d the vowels and the consonants alternate,

e the vowels are together.

11 The digits 3, 3, 4, 4, 4 and 5 are placed in a row to form a six-digit number. If one of these numbers is selected at random, find the probability that:

a it is even,

b it ends in 5,

c the 4s occur together,

d the number starts with 5 and then the 4s and 3s alternate,

e the 3s are separated by at least one other number.

12 The letters of the word PRINTER are arranged in a row. Find the probability that:

a the word starts with the letter E,

b the letters I and P are next to one another,

c there are three letters between N and T,

d there are at least three letters between N and T.

13 The letters of KETTLE are arranged randomly in a row. Find the probability that:

a the two letters E are together,

b the two letters E are not together,

c the two letters E are together and the two letters T are together,

d the Es and Ts are together in one group.

14 The letters of ENTERTAINMENT are arranged in a row. Find the probability that:

a the letters E are together,

b two Es are together and one is apart,

c all the letters E are apart,

d the word starts and ends with E.

15 A tank contains 20 tagged fish and 80 untagged fish. On each day, four fish are selected at random, and after noting whether they are tagged or untagged, they are returned to the tank. Answer the following questions, correct to three significant figures.

a What is the probability of selecting no tagged fish on a given day?

b What is the probability of selecting at least one tagged fish on a given day?

c Calculate the probability of selecting no tagged fish on every day for a week.

d What is the probability of selecting no tagged fish on exactly three of the seven days during the week?

16 A bag contains seven white and five black discs. Three discs are chosen from the bag. Find the probability that all three discs are black, if the discs are chosen:

a without replacement,

b with replacement,

c so that after each draw the disc is replaced with one of the opposite colour.

17 Six people are to be divided into two groups, each with at least one person. Find the probability that:

a there will be three in each group,

b there will be two in one group and four in the other,

c there will be one group of five and an individual.

18 A three-digit number is formed from the digits 3, 4, 5, 6 and 7 (no repetitions allowed). Find the probability that:

a the number is 473,

b the number is odd,

c the number is divisible by 5,

d the number is divisible by 3,

e the number starts with 4 and ends with 7,

f the number contains the digit 3,

g the number contains the digits 3 and 5,

h the number contains the digit 3 or 5,

i all digits in the number are odd,

j the number is greater than 500.

19 The digits 1, 2, 3 and 4 are used to form numbers that may have 1, 2, 3 or 4 digits in them. If one of the numbers is selected at random, find the probability that:

a it has three digits,

b it is even,

c it is greater than 200,

d it is odd and greater than 3000,

e it is divisible by 3.

20 **a** A senate committee of five members is to be selected from six Labor and five Liberal senators. What is the probability that Labor will have a majority on the committee?

b The senate committee is to be selected from N Labor and five Liberal senators. Use trial and error to find the minimum value of N, given that the probability of Labor having a majority on the committee is greater than $\frac{3}{4}$.

21 Four basketball teams A, B, C and D each consist of ten players, and in each team, the players are numbered 1, 2, ..., 9, 10. Five players are to be selected at random from the four teams. Find the probability that of the five players selected:

a three are numbered 4 and two are numbered 9,

b at least four are from the same team.

22 A poker hand of five cards is dealt from a standard pack of 52. Find the probability of obtaining:

a one pair,

b two pairs,

c three of a kind,

d four of a kind,

e a full house (one pair and three of a kind),

f a straight (five cards in sequence regardless of suit, ace high or low),

g a flush (five cards of the same suit),

h a royal flush (ten, jack, queen, king and ace in a single suit).

23 Four adults are standing in a room that has five exits. Each adult is equally likely to leave the room through any one of the five exits.

a What is the probability that all four adults leave the room via the same exit?

b What is the probability that three particular adults use the same exit and the fourth adult uses a different exit?

c What is the probability that any three of the four adults come out the same exit, and the remaining adult comes out a different exit?

d What is the probability that no more than two adults come out any one exit?

24 **a** Five diners in a restaurant choose randomly from a menu featuring five main courses. Find the probability that exactly one of the main courses is not chosen by any of the diners.

b Repeat the question if there are n diners and a choice of n main courses.

25 [The birthday problem]

a Assuming a 365-day year, find the probability that in a group of three people there will be at least one birthday in common. Answer correct to two significant figures.

b If there are n people in the group, find an expression for the probability of at least one common birthday.

c By choosing a number of values of n, plot a graph of the probability of at least one common birthday against n for $n \leq 50$.

d How many people need to be in the group before the probability exceeds 0.5?

e How many people need to be in the group before the probability exceeds 90%?

ENRICHMENT

26 During the seven games of the football season, Max and Bert must each miss three consecutive games. The games to be missed by each player are randomly and independently selected.

a What is the probability that they both have the first game off together?

b What is the probability that the second game is the first one where Max and Bert are both missing?

c What is the probability that Max and Bert miss at least one of the same games?

27 Eight players make the quarter-finals at Wimbledon. The winner of each of the quarter-finals plays a semi-final to see who enters the final.

a Assuming that all eight players are equally likely to win a match, show that the probability that any two particular players will play each other is $\frac{1}{4}$.

b What is the probability that two particular people will play each other if the tournament starts with 16 players?

c What is the probability that two particular players will meet in a similar knockout tournament if 2^n players enter?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **15 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q8** 1 mark

A bag contains 5 identical blue marbles, 6 identical black marbles and 3 identical red marbles. Three marbles are drawn at random.
Which expression below gives the correct probability that exactly two blue marbles are drawn?

(A) $\dfrac{{}^5C_2}{{}^{14}C_3}$

(B) $\dfrac{{}^5C_2 \times {}^9C_1}{{}^{14}C_3}$

(C) $\dfrac{2}{5} \times \dfrac{1}{9}$

(D) $\dfrac{2}{14} \times \dfrac{1}{9}$

Hint: is the selection of three marbles ordered or unordered?

PP 2 **Killara High School, 2023, Year 12 Examination, Mathematics Extension 1 Trial, Q6** 1 mark

A lottery has 30 numbers and 6 are drawn without replacement. A player of this game chooses 6 distinct numbers themselves.
The probability of matching at least 4 of the 6 numbers is given by:

A. 0.005

B. 0.007

C. 0.018

D. 0.046

Hint: which numbers of matches count as 'at least 4'?

PP 3 **Normanhurst Boys High School, 2020, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(f)** 2 marks

At a soccer club a team of 13 players is to be chosen from a pool of 25 players consisting of 20 female players and 5 male players.
What is the probability that the team will consist of only female players?

Hint: what is the sample space, and is a team ordered or unordered?

PP 4 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q11(c)** 2 marks

2 cards are chosen at random, without replacement, from a standard 52-deck of playing cards. Find the probability of choosing at least 1 heart.

Hint: 'at least 1' is a warning word.

PP 5 **Hornsby Girls High School, 2022, Mathematics Extension 1, Year 12 Higher School Certificate Trial Examination Term 3, Q12(c)** 2 marks

A class consists of 10 girls and 8 boys. What is the probability of selecting a committee of 5, that contains 3 girls and 2 boys.

Hint: think about the girls and the boys separately.

PP 6 **Normanhurst Boys High School, 2022, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q11(c)(i)(ii)** 3 marks

A school has twenty prefects. Jun and Alex are two of those prefects.
A committee of seven prefects is to be formed to organise a function.

(i) How many different committees can be formed? 1

(ii) What is the probability that the committee contains Jun but does not contain Alex? 2

Hint: each committee of seven is equally likely to be chosen.

PP 7 **Girraween High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q11(c)(i)(ii)** 4 marks

A group of 40 new Year 7 students is going to be randomly divided into two classes A and B of 20 students each. Four of the students, Anne, Ben, Carrie and David have been together in the same class since they started school and are close friends.

(i) What is the probability that the four friends will be in the same class? 2

(ii) What is the probability that exactly three of the friends will be in the same class? 2

Hint: think about where the four friends can end up.

CHAPTER 14 REVIEW

Chapter review exercise

Mixed practice from the whole chapter, Lessons 14A to 14F.

1 How many ways can 8 people line up in a queue?

2 By unrolling the factorial (see Box 2 in Lesson 14A) simplify:

a $\dfrac{9!}{7!}$

b $\dfrac{(n+1)!}{(n-1)!}$

c $(k+1)! - k!$

3 Use your calculator to evaluate:

a ${}^{12}\mathrm{C}_{7}$

b ${}^{10}\mathrm{C}_{3} \div {}^{6}\mathrm{C}_{3}$

c $\binom{20}{17}$

4 How many four-letter words, with no repeated letters, can be formed from the letters of JACKSON?

5 A number plate in a certain country consists of 3 letters A–Z followed by 4 digits 0–9. How many number plates are possible?

6 Find how many arrangements of the letters of the word FOUNDER are possible:

a if the vowels and consonants must alternate,

b if the word must start with N and end with D,

c if all the consonants must be in a group at the end of the word,

d if the R is somewhere to the right of the U.

7 Five boys and five girls are to sit in a row. Find how many ways this can be done if:

a there are no restrictions,

b the boys and girls sit in distinct groups,

c a particular boy and girl must sit together.

8 How many ways can the letters of REPORTER be arranged?

9 How many words of three or four letters may be formed using the letters of SAMUEL?

10 A quiz consists of ten questions, each taking the answer Yes or No. How many ways is it possible to get 6 correct and 4 wrong answers?

11 A committee of seven is to be chosen from six men and ten women. Find how many committees are possible if:

a there are no restrictions,

b all members are to be female,

c all members are to be male,

d there are to be exactly two men,

e there are to be four women and three men,

f there is to be a majority of women,

g a particular man must be included,

h a particular man must not be included,

i Mustafa refuses to be on a committee with Ying Yue.

12 Eight people arrive at a restaurant. Find how many ways can they be assigned to:

a a large table for five and a smaller table for three,

b two quite different tables for four,

c two indistinguishable tables for four.

13 What is the probability that if a committee of six is formed at random from four men and three women, it will have more men than women?

14 From a standard pack of 52 cards, three are selected at random. Find the probability that:

a they are the queen of spades, the three of clubs and the nine of hearts,

b all three are kings,

c they are all clubs,

d they are all of the same suit,

e one is red and two are black,

f one is a three, one is a five, and one is an eight,

g two are fives and one is a seven,

h at least two of them are spades.